

Subordinates in Charge:

Does Delegation Improve Bank Supervision?

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Abstract

We develop a model of bias and information loss in supervisory communication and apply it to evaluate a policy reform that delegated supervisory decision-making authority over a subset of bank branches to a lower level. Affected branches become 57-80% more likely to face supervisory intervention, implying substantial efficiency gains arising from improved detection of banking misconduct and more accurate assessments of its severity. The evidence is inconsistent with alternative explanations, including increased supervisory stringency or greater risk-taking by banks. Our analysis highlights decentralization benefits within supervisory hierarchies specifically, and speaks to the optimal organizational design for fraud detection more broadly.

Keywords: Communication, decentralization, financial architecture, fraud, organizational design, supervision

JEL: D23, G21, G28

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1 Introduction

A central question in the design of financial architecture is how supervisory powers should be allocated between local and central authorities. Much of the debate focuses on settings in which local supervisors are *independent* of the central authority and finds that local decision-making produces less stringent supervisory outcomes.¹ This is commonly attributed to local supervisors being captured by local interests or failing to internalize spillovers across regions.

This paper shows that within a supervisory *hierarchy*, local supervision may deliver both more stringent and more efficient outcomes.² The reason is that divergent interests of local and central supervisors make communication between them less informative. This results in a loss of local information when decisions are made at the central level and prevents desirable interventions from being undertaken. Delegating authority to (biased) local supervisors can therefore make supervision both more stringent and more efficient.

We examine a decentralization reform in China and interpret the findings through the lens of a model of supervisory communication and bias. In 2015, China decentralized supervision of branches belonging to banks classified as “local” but not of branches belonging to “national” banks.³ Prior to 2015, all branches were supervised in a hub-and-spoke system in which information was predominantly collected by supervisors at the prefecture-level, while decisions involved the central supervisor, requiring communication between the two. The reform fully delegated authority over branches of local banks to local supervisors.

We use a large, partly hand-collected, dataset covering 5,429 prefecture-level branches over a ten-year window around the 2015 policy reform. We measure supervisory outcomes through formal

¹See, for example, [Agarwal et al. \(2014\)](#), [Haselmann et al. \(2022\)](#) and [Beck et al. \(2023\)](#). Note that less stringent supervision does not necessarily imply worse supervision. Theory offers divergent predictions on whether centralized or decentralized settings are preferred ([Dell’Ariccia and Marquez, 2006](#); [Calzolari et al., 2019](#); [Lóránth et al., 2022](#)). See [Ampudia et al. \(2019\)](#) for an overview of the policy debate.

²Hierarchical, or hub-and-spoke, supervisory structures are common. In the United States, the Federal Reserve System (Fed), the Federal Deposit Insurance Corporation (FDIC), and the Office of the Comptroller of the Currency (OCC) operate through headquarters and regional offices. Similar distributed structures also exist in Germany’s Deutsche Bundesbank (Bundesbank), India’s Reserve Bank of India (RBI), and Canada’s Office of the Superintendent of Financial Institutions (OSFI).

³The Chinese banking sector is the largest in the world with about \$40 trillion in assets. It serves more than 800 million individuals through more than 4,000 commercial banks.

declarations of banking misconduct (“supervisory interventions”). To do so, we assemble novel data at both the branch- and supervisory-office-level on these interventions.⁴ The granularity of the data allows us to exploit variation across branches within the same bank as well as within a supervisory office, comparing decisions involving branches of different banks.

We find that branches of local banks are more likely to face a supervisory intervention following the reform. Specifically, the probability that a local branch receives a supervisory intervention increases by 6.3 to 8.8 percentage points (pp) relative to branches of a national bank, which corresponds to 57 to 80% of the unconditional probability of intervention. The result is robust along a wide set of specifications, including allowing the effect of the reform to vary with bank characteristics.

What explains why decentralization leads to more interventions, even though local supervisors are usually considered more lenient? We present a model in which local supervisors are better informed about potential misconduct but are biased against interventions. This bias impairs communication (modeled as cheap talk) between local and central supervisors under centralization. Specifically, the local supervisor wants to avoid strict intervention by the central supervisor when the level of misconduct is moderate. This results in the local supervisor not reporting such incidents to the central supervisor, thereby deterring desirable supervisory intervention. Decentralization restores information in supervisory decision-making and can lead to more detection of misconduct even though the local supervisor is biased against interventions.

While our model is agnostic about whether decentralization is overall desirable,⁵ it is always desirable in the parameter range where supervisory interventions increase. The implied welfare improvements are material. In particular, our baseline estimate suggests that the reform eliminates 19% of the efficiency losses associated with centralized supervision.

Our model has two important implications. First, moderate levels of misconduct are now also detected. Such misconduct warrants lower penalties, so the average fine issued in a supervisory

⁴In the Chinese system, a supervisory intervention is a formal measure taken when a bank is suspected of violating rules or prudent practice. It identifies the specific violation, mandates corrective action, and imposes a penalty. More than 300 local supervisory offices gained greater decision-making authority as a result of the reform.

⁵If the importance of local relative to public information is modest, or if the local bias is very large, centralization is in fact preferred.

intervention declines. Second, fines become more dispersed. This is because the local supervisor uses granular information when deciding about the severity of an intervention, resulting in penalties that are responsive to the underlying misconduct. Both predictions are confirmed in the data. Our model also identifies two factors that determine the extent to which decentralization improves outcomes: central access to local information and the predictability of that information. Using proxies for both, we find that the reform’s effect is stronger when (incremental) local information is more valuable.

Our findings on the frequency, severity, and dispersion of supervisory interventions, as well as on the value of local information, speak to an informational interpretation of the reform. They are difficult to reconcile with alternative explanations. First, there may have been a confounding shock around the time of the reform that increased risk specifically for branches of local banks. While such a shock could generate more interventions in local branches, it should also increase average fines, contrary to our findings. The risk explanation is also inconsistent with evidence, discussed below, that local banks became more risk-averse following tighter supervision. Second, the reform may have changed local supervisors’ preferences for intervening in local banks. But if supervisors had simply become stricter towards local banks, one would again expect average fines to rise rather than fall. Third, supervisory capacity may have increased after the reform, potentially explaining why supervisory outcomes improve. However, our results are robust to controlling for changes in supervisory capacity, proxied by new hires.

We conclude by examining whether improved supervision reduces bank risk-taking. Using data on individual lending decisions, we find that branches of local banks become more conservative following reform: they require higher compensation for risk and reduce loan sizes, thereby taking on less credit risk. At the aggregate level, prefectures with a higher share of branches of local banks experience lower loan supply following the reform.

Our paper has an important message for policy. A large part of the discussion on the design of supervisory architecture, especially in the Eurozone, has centered on the centralization of supervision relative to independent supervisors ([Ampudia et al., 2019](#)), with evidence pointing to stricter outcomes under centralization. For example, [Agarwal et al. \(2014\)](#) show that federal regulators are

systematically tougher than state regulators, while [Haselmann et al. \(2022\)](#) find that large significant banks face higher standards under the Single Supervisory Mechanism (SSM).⁶ Our analysis shows that in hierarchical supervisory systems of the kind observed in many countries, outcomes may be stricter (and more efficient) when the local supervisor is granted authority. The key mechanism we highlight is that local bias distorts communication upward in the hierarchy, making the central supervisor less informed about local conditions⁷ and, as a result, intervene less.⁸

Our paper is related to a growing theoretical literature that analyzes optimal financial architectures involving multiple supervisors and regulators. A large part of this literature has studied the trade-off between a setting with a single central regulator and one with several independent local regulators ([Dell’Ariccia and Marquez, 2006](#); [Kara, 2016](#); [Foarta, 2018](#); [Calzolari et al., 2019](#); [Bolton and Oehmke, 2019](#); [Colliard, 2020](#); [Lóránth et al., 2022](#)). This literature has primarily focused on inefficiencies arising from externalities across jurisdictions. [Repullo \(2018\)](#) and [Carletti et al. \(2021\)](#) study the incentives of local and central supervisors to acquire information within a hierarchy. Our paper instead focuses on the role of coarse communication ([Crawford and Sobel, 1982](#)) in the optimal design of supervision. Such communication has also been shown to shape outcomes within firms, in terms of corporate governance ([Harris and Raviv, 2008](#); [Malenko, 2014](#); [Baldenius et al., 2014](#); [Malenko, 2024](#)) and interactions among different layers of management ([Harris and Raviv, 2005](#); [Grenadier et al., 2016](#); [Chakraborty and Yilmaz, 2017](#); [Levit, 2020](#)).

Our model builds on [Dessein \(2002\)](#) who analyzes the benefits of delegation relative to communication and finds that there is a trade-off between information and incentives but that delegation is preferred over a wide range of parameter values.⁹ Our work extends [Dessein \(2002\)](#) to a setting with both an extensive and an intensive margin (the decision to intervene and the severity of inter-

⁶Our model shows that stricter supervision should not be equated with greater efficiency. Decentralization may be optimal even if it produces fewer interventions, because the interventions that do occur better reflect underlying misconduct.

⁷Several papers highlight the importance of local information in the supervisory process. For example, [Lim et al. \(2023\)](#) and [Gopalan et al. \(2021\)](#) show that closures of local offices of central supervisors are associated with greater bank risk.

⁸Another reason why, in practice, decentralization to subordinated (relative to independent) supervisors is more beneficial is that, within a hierarchy, a bias of the local supervisor is more easily contained (as the latter is accountable to the central supervisor).

⁹In contrast to [Dessein \(2002\)](#), in [Aghion and Tirole \(1997\)](#) information acquisition is endogenous and affected by decentralization. Other papers that consider decentralization in a general principal-agent context include [Melumad and Reichelstein \(1987\)](#), [Bolton and Dewatripont \(1994\)](#) and [Alonso et al. \(2008\)](#).

vention). We show that due to the presence of the extensive margin, the overall trade-off disappears – that is, interventions can become both more informed and less biased. The model also maps into observable outcomes, allowing us to infer the efficiency implications of an actual decentralization reform.

Beyond supervision, the insights from our model apply more broadly to the question of which organizational level should have the power to discipline when misconduct or fraud is suspected. Lower-level managers have incentives to withhold information about misconduct for fear of punishment (of their subordinates, or themselves). While this seemingly favours making discipline a central function, our model shows a cost to that: Lower-level managers may fail to report moderate misconduct, and withhold information about the severity of larger misconduct.¹⁰ Our model also applies to the optimal allocation of authority over investment decisions in case lower-level managers are biased against undertaking projects, and against larger projects.¹¹

Empirical studies have emphasized benefits, but also costs, of decentralization in several contexts. [Li \(2017\)](#) shows that experts are more biased but still make better decisions because they possess better information. [Aghion et al. \(2021\)](#) show that firms that delegated greater decision-making authority to local plant managers prior to the Great Recession outperformed their more centralized counterparts in sectors most severely affected by the subsequent crisis. [Bandiera et al. \(2021\)](#) analyze the consequences of granting autonomy of procurement officers in Pakistan, and find that autonomy reduces procurement prices without reducing quality. [Kala \(2024\)](#) finds that granting managers in state-owned enterprises more autonomy results in greater value added but reduces investment in activities that primarily generate social benefits. [Limodio et al. \(2026\)](#) find that ethnic conflict in Ethiopia increases the value of local information, leading banks to appoint locally matched managers but reduce their decision-making autonomy. [Vannutelli \(2026\)](#) shows that reducing local control over auditor appointments strengthens alignment between municipal fiscal behavior and national fiscal goals.

¹⁰The ability of organizational structure to influence risk in institutions has received considerable attention in the context of banks ([Laeven and Levine, 2009](#); [Fahlenbrach and Stulz, 2011](#); [Beltratti and Stulz, 2012](#); [Ellul and Yerramilli, 2013](#)) and firms more broadly ([Burns and Kedia, 2006](#); [Efendi et al., 2007](#); [Khanna et al., 2015](#)).

¹¹Such bias may arise because undertaking new projects requires effort and exposes the manager to the risk of failure.

2 Institutional Background

2.1 An overview of China’s banking sector

The Chinese banking system has experienced substantial growth and is now the world’s largest. At the end of our sample period (2020) it had \$38.98 trillion assets, compared to \$27.71 trillion in the United States. There are over 4,000 commercial banks. Eighteen of them operate nationwide, including the 6 largest state-owned banks and the 12 national joint-stock banks. These national banks collectively held \$26.96 trillion assets at the end of 2020, accounting for 69.5% of all commercial bank assets. In addition to the national banks, there is a range of regional institutions that we refer to as local banks: 134 prefecture-level commercial banks; approximately 1,600 rural commercial banks; several hundred of rural credit cooperatives; and numerous village banks.

Commercial banks are predominantly organized through branches. The typical organizational structure consists of a headquarter, a branch (“Fenhang” in Chinese) in each prefecture where the bank is active, numerous lower-level offices (“Zhihang”), and even more local branches. Our analysis will focus on the prefecture-level branch, which is responsible for all banking activities in the prefecture, including the offices and the local branches. A prefecture is an administrative unit similar to a Metropolitan Statistical Area in the United States and has a median population of more than 3.3 million (larger than the median US state).

2.2 Regulatory and supervisory framework

The China Banking Regulatory Commission (CBRC) was created in 2003 as the main authority regulating and supervising the Chinese banking sector. In 2018, the CBRC merged with the China Insurance Regulatory Commission (CIRC), to become the China Banking and Insurance Regulatory Commission (CBIRC).¹² The CBRC has a hub-and-spoke structure similar to the one present in many other countries (see footnote 2). Headquartered in Beijing, the CBRC supervises all commercial banks through a network of local supervisory offices. This network comprises provincial

¹²In 2023, after the end of our sample period, the National Financial Regulatory Administration (NFRA) was created, taking over supervisory responsibilities and personnel from the CBRC.

offices (CBRC bureaus) in the capitals of the 31 provinces and in five major metropolitan areas (Dalian, Ningbo, Xiamen, Qingdao, and Shenzhen), and municipal offices (CBRC sub-bureaus) in 306 prefectures (see Figure IA1 in the Internet Appendix). Comparable to the field offices of the OCC in the United States, these offices (bureaus and sub-bureaus) serve as local entities overseeing the banks within their respective jurisdictions.

The CBRC has a fully hierarchical management structure. The CBRC’s head (or central) office in Beijing establishes rules, guidelines, and policies. It also directly appoints the heads of local offices. There is, in principle, full alignment of the objectives between central and local supervision.

Supervision is primarily implemented through on-site inspections. Banks undergo periodic inspections at least once every five years, although many are inspected more frequently, with some supervisors conducting inspections on a one- to two-year rotational basis. Supervisors also formulate annual thematic inspection plans based on a combination of random sampling and risk-based supervision, while ad hoc inspections may be initiated in response to whistleblower reports, public complaints, or other intelligence.

2.3 The decentralization reform of 2015

In January 2015, the CBRC implemented its first major organizational reform. In the area of supervision, the primary objective is to decentralize administrative powers, bringing supervisors closer to financial institutions. Specifically, the reform transfers the supervisory responsibilities and powers for local banks to local supervisors, without changing the organization of supervision for national banks.

Prior to reform, the central office and local offices of CBRC jointly supervised all banks,¹³ with the ultimate authority being with the central supervisor. The most common case regarding supervisory intervention (and resulting penalty) is that local offices provided recommendations to the central office, which then made the final decision. In other cases, local offices decided on an intervention first and informed the central office, possibly leading to an overruling of the lower-level decision. In a few cases, investigations by a local office were also joined by staff from the central

¹³With the exception of the headquarters of the national banks, which are solely supervised by the central office.

office, and joint decisions were reached. The reform fully transferred supervision for local banks, and specifically decisions on interventions, to the local offices.¹⁴ The local offices now independently decide on interventions, without an obligation to communicate about individual interventions the central office. While the reform makes supervision of local banks the sole task of local supervisors, it does not modify who sets the overall objectives of supervision. Those remain in the full realm of the CBRC's head office, to which the local offices are accountable.

3 A Model of Supervisory Communication and Bias

3.1 Overview

The relation between central and local supervisors in China resembles a principal-agent setting, as the the latter is subordinated to the former who sets the overall objectives of supervision. Prior to the 2015 reform, the central supervisor had ultimate authority over decisions (even though supervision was carried out jointly). Information about the banks was primarily in the hand of the local supervisor, who communicated this information to the central supervisor (either directly through a recommendation, or indirectly through an initial decision). The reform then fully delegated decision-making authority to the local supervisor.

We develop a model that allows analyzing the impact of the reform on supervisory decision-making.¹⁵ The main tension in the model is that the local supervisor has better information about banks, but faces distorted incentives as he is biased against interventions. A central insight of the analysis is that local bias substantially distorts decision-making also under centralized decision-making. The reason is that the local supervisor communicates strategically to the central supervisor, in order to influence the latter's decisions. This results in two costs. First, also decisions by the central supervisor are subject to a bias (that goes in the same direction as the local supervisor's bias). Second, as the local supervisor does not fully transmit information, there is an informational

¹⁴See Articles 13 and 14 of the 2015 Interim Measures for On-site Inspections of the China Banking Regulatory Commission (www.gov.cn/gongbao/content/2016/content_5041571.htm; last accessed: June 2026) and the 2015 Measures of the China Banking Regulatory Commission on Administrative Penalties (www.gov.cn/gongbao/content/2015/content_2973203.htm; last accessed: June 2026).

¹⁵The model adapts Dessein (2002) to a setting in which the principal makes two decisions that can be observed in the data: the intervention decision (extensive margin) and the size of the fine (intensive margin).

loss, making the severity of interventions less tailored to the severity of misconduct.

The model shows that delegation can be both welfare enhancing and reducing, and derives conditions for either case. Based on our empirical estimates this allows to exclude the case where delegation lowers welfare. The model also allows us to derive a range of empirical predictions that can be contrasted against alternative explanations for why supervisory interventions may have changed following the reform.

3.2 Setup

There is a central supervisor, also called the principal (she), and a local supervisor, also called the agent (he). The type of a bank is given by $\theta + \delta$. The component θ is observed only by the agent and is independently and uniformly distributed on $[-1, 1]$. The component δ is public information and is observed by both principal and agent. It takes values

$$\delta \in \{-\bar{\delta}, \bar{\delta}\}$$

with equal probability, where $\bar{\delta} \in (0, \frac{1}{2})$ measures the importance of public information.¹⁶ The bank type can be interpreted as the likelihood that the bank has violated supervisory guidelines, the severity of the violation, or the bank's overall risk-taking.

The central supervisor would like to impose a fine whenever $\theta + \delta > 0$. Her ideal fine is

$$F^*(\theta, \delta) = \max\{0, \theta + \delta\}.$$

The local supervisor is more lenient. He would like to impose a fine only when $\theta + \delta > b$, where $b > 0$. His ideal fine is

$$F_A^*(\theta, \delta) = \max(0, \theta + \delta - b).$$

Preferences over fines (as illustrated in Figure 1) are represented by absolute-loss utility. If the actual fine is $F \geq 0$, the principal's pay-off is $U_P(F, \theta, \delta) = -|F - F^*(\theta, \delta)|$ and the agent's

¹⁶The restriction $\bar{\delta} < 1/2$ reduces the number of cases that must be discussed and simplifies the exposition.

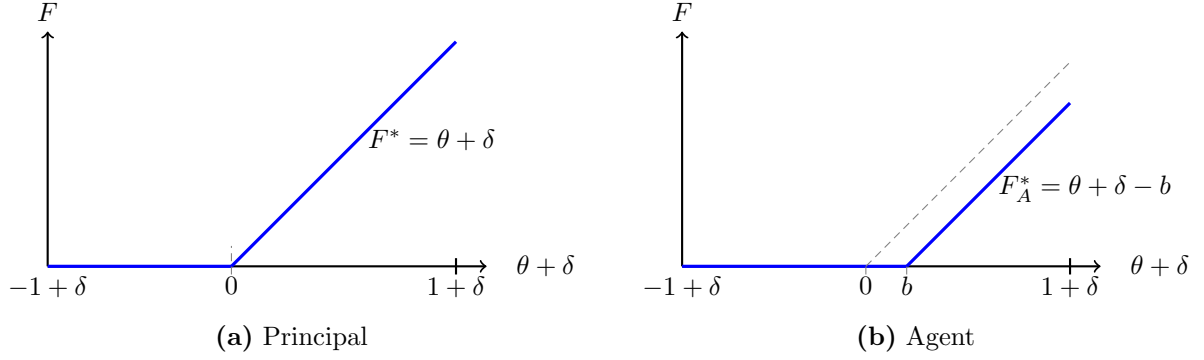


Figure 1: Ideal Fines.

Panel A depicts the ideal fine of the principal. Panel B depicts the ideal fine of the agent.

payoff is $U_A(F, \theta, \delta) = -|F - F_A^*(\theta, \delta)|$. We refer to an outcome with a positive fine, $F > 0$, as a supervisory intervention.

3.3 Organization design

We consider two institutional settings.

First, under decentralized supervision, final authority is delegated to the local supervisor. The agent then chooses his ideal fine, so the realized fine is

$$F = \max\{0, \theta + \delta - b\}.$$

Second, under centralized supervision, the central supervisor chooses the fine, but she can consult the local supervisor. Communication is modeled as cheap talk ([Crawford and Sobel, 1982](#); [Dessein, 2002](#)). The agent sends a message m , and the principal chooses a fine

$$F(m) \in \arg \max_{F \geq 0} E[U_P(F, \theta, \delta) | G(\theta, m)]$$

where the conditional distribution of θ after message m is the determined in equilibrium.

3.4 Equilibrium under centralization

We now characterize the cheap-talk outcome under centralization. As in [Crawford and Sobel \(1982\)](#), all equilibria are characterized by a partition $\{a_0, a_1, a_2, \dots, a_n\}$ of $[-1, 1]$, with $a_0 = -1$ and $a_n = 1$, and are outcome-equivalent to an equilibrium with n messages $m_k \in M = \{m_1, \dots, m_n\}$ in which

- (i) the agent sends message m_k whenever $\theta \in (a_{k-1}, a_k]$, so that $G(\theta|m_k) = U(a_{k-1}, a_k]$,
- (ii) upon receiving the message m_k , the principal issues a fine

$$F_k = \max\{0, \delta + \frac{a_{k-1} + a_k}{2}\} > 0,$$

Note that for this to be an equilibrium, we must have that at $\theta = a_k$, the agent is indifferent between fines F_k and F_{k+1} whenever $1 < k < n$.

As is standard, we focus on the most informative cheap talk equilibrium (see, e.g., [Dessein, 2002](#)), since this equilibrium is ex-ante preferred by both principal and agent. As is well known, there always exists an equilibrium in which $n = 1$ and any communication is uninformative.

For a fixed value of δ , define

$$n^* = \left\lfloor 1 + \sqrt{\frac{1 + \delta}{2b}} \right\rfloor$$

where $\lfloor x \rfloor$ denotes the largest integer below x .

Proposition 1. (i) When $E(\theta + \delta) < 0$ (because $\delta = -\bar{\delta}$):

- If $b > \frac{1}{2} - \frac{\bar{\delta}}{2}$, there is no informative communication and $F = 0$.
- If $b < \frac{1}{2} - \frac{\bar{\delta}}{2}$, there is informative communication with $n^* > 1$ messages,
 - the principal issues no fine, $F_1 = 0$, upon receiving the message m_1 ,
 - the principal issues a fine $F_k > 0$ upon receiving message m_k with $k > 1$,
 - the agent sends m_1 if and only if $\theta < a_1$ where

$$a_1 = (n^* \cdot b - \delta) \cdot \frac{2(n^* - 1)}{2n^* - 1} + \frac{1}{2n^* - 1} > b - \delta \tag{1}$$

(ii) When $E(\theta + \delta) > 0$ (because $\delta = \bar{\delta}$):

- If $b > 1/2$, there is no informative communication and $F = \bar{\delta} > 0$.
- If $b \in (\bar{b}, 1/2)$ with $\bar{b} \equiv \max\{\frac{1}{2} - \bar{\delta}, \frac{1}{3} - \frac{\bar{\delta}}{2}\}$, there is informative communication but all messages induce strictly positive fines ($F_k > 0$ for any k).
- If $b < \bar{b}$ there is informative communication with a no fine message: $F_1 = 0$ following message m_1 and $F_k > 0$ following message m_k with $k > 1$. As above, the equilibrium has n^* messages and a_1 is given by (1).

Proof. See Appendix A. □

The above proposition shows that for moderate biases ($b < \frac{1}{2} - \frac{\bar{\delta}}{2}$ or $b < \bar{b}$), informative communication between principal and agent is possible and the principal does not issue a fine whenever the agent recommends her not to do so (i.e. whenever the agent sends message m_1). In this range, as long as $b > \frac{1+\delta}{8}$, the agent can only indicate whether the principal should issue a strictly positive fine or not ($n^* = 2$). Only for b very small, that is $b < \frac{1+\delta}{8}$, the agent may be able to indicate whether a desired fine should be small or large, or even finer gradations ($n^* \geq 3$).

For large biases, that is when $b > \min\{\frac{1}{2} + \frac{\delta}{2}, \frac{1}{2}\}$, no informative communication is feasible and intervention depends on the sign of δ .

Finally, when $\delta > 0$, there is also small parameter range when $b \in (\bar{b}, \frac{1}{2})$, where informative communication is feasible, but the principal always intervenes.

3.5 Optimality of Delegation and Likelihood of Supervisory Interventions

We now examine the circumstances under which delegation is preferred to centralization from the principal's perspective, and the corresponding implications for observable supervisory outcomes. Supervisory outcomes under delegation are given by $F = \max\{0, \theta + \delta - b\}$, while outcomes under centralization are $F = \max\{0, E(\theta + \delta | m_i)\}$.

Proposition 2. *The optimality of delegation and the relative frequency of interventions depend critically on the severity of the local supervisor's bias, b :*

- (i) **High Bias** ($b > 1 - (\sqrt{2} - 1)\bar{\delta}$): *Centralization is optimal, and interventions are more frequent under centralization than under delegation.*

(ii) **Medium Bias** ($\bar{b} < b < 1 - (\sqrt{2} - 1)\bar{\delta}$): *Delegation is optimal, but interventions remain more frequent under centralization.*

(iii) **Low Bias** ($b < \bar{b}$): *Delegation is optimal, and interventions are strictly more frequent under delegation. Under delegation, there is no intervention when $\theta < b - \delta$. Under centralization, there is no intervention when $\theta < a_1$, where $a_1 > b - \delta$.*

Proof. See Appendix A. □

To unpack the intuition behind these results, consider first the case where the bias is sufficiently large so that the agent never intervenes. The principal, observing a positive public signal ($\delta = \bar{\delta} > 0$), optimally intervenes under centralization, making centralization both strictly preferred and more interventionist.

As the bias shrinks to an intermediate range (b close to $1/2$), the agent intervenes when $\theta + \delta > b$ and delegation becomes optimal due to its informational benefits: fines are responsive to the local state θ : $F = \theta + \delta - b$. In contrast, the principal indiscriminately intervenes whenever $\delta > 0$ setting $F = \bar{\delta}$, independently of the realization of θ as communication is un-informative (Proposition 1). Centralization still yields a higher sheer volume of interventions, however, as the principal intervenes in at least half of all cases, compared to $(1 - b)/2$ of cases when the agent sets fines.

Note that the above logic in favor of delegation is more subtle when $\bar{\delta}$ is large, as then there is a substantial benefit from intervening under centralization when $\delta = \bar{\delta} > 0$. A trade-off then arises between more, larger but also rather indiscriminate interventions under centralization (always intervene when $\delta > 0$), and less frequent, smaller interventions under delegation that are tailored to the value of θ (intervene when $\theta + \delta > b$). As we show in the Appendix A, this trade-off resolves in favor of delegation when b is not too large relative to $\bar{\delta}$, that is when $b < 1 - (\sqrt{2} - 1)\bar{\delta}$. Intuitively, as public information becomes more relevant (a larger $\bar{\delta}$), delegation is less likely to be optimal.

The most nuanced — and empirically relevant — case emerges when the bias is small ($b < \bar{b}$). In this regime, delegation not only remains strictly optimal but also paradoxically results in *more* frequent interventions than centralization, despite the local supervisor being inherently more lenient than the principal.

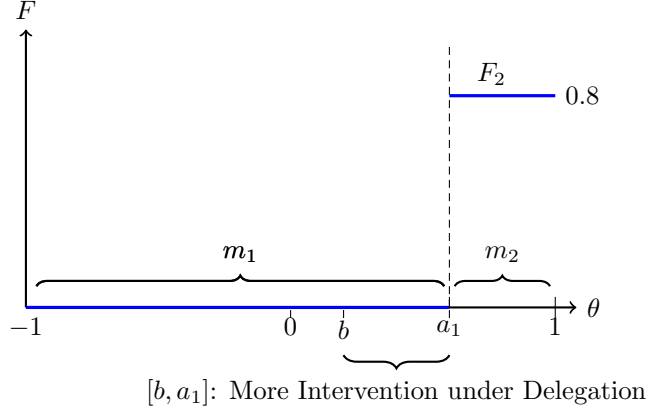


Figure 2: More intervention under delegation.

Example with $\bar{\delta} = 0$ and $b = 0.2$. For types $\theta \in (b - \delta, a_1)$, the agent desires a small fine but sends message m_1 (yielding $F_1 = 0$) to prevent the principal from imposing a large fine ($F_2 \gg 0$).

This counterintuitive result hinges on strategic communication. Because the agent’s bias is small, the principal defers to the agent when he sends the lowest-risk message (m_1), meaning the agent wields “real authority” (Aghion and Tirole, 1997) over the extensive margin. However, the agent strategically expands the use of this zero-fine message to shield banks with moderate violations. Specifically, for intermediate realizations of the bank type where $\theta \in (b - \delta, a_1)$, the local supervisor observes a violation and *would* issue a small, tailored fine under delegation. But under centralization, revealing this violation would trigger the principal to impose a severe, uncalibrated fine. To prevent this disproportionate punishment, the agent conceals the information entirely, recommending a zero fine (m_1). Figure 2 illustrates this strategic omission. By delegating authority, the principal eliminates the agent’s fear of excessive punishment, resulting in interventions for these moderate violations (or types).

Consequently, delegation strictly improves efficiency through both the intensive and extensive margins. On the intensive margin, there is a recognized trade-off between fines that are too small (due to the agent’s bias) but perfectly tailored to θ , mirroring the classic insights of Dessein (2002). On the extensive margin, however, delegation provides an unambiguous efficiency gain: it strictly increases the number of warranted interventions. As we demonstrate in Section 3.6.1, this extensive margin effect dominates. Overall, delegation increases the total volume of appropriate fines issued.

Empirical mapping This theoretical framework provides a direct lens for interpreting our empirical results. Proposition 2 establishes that whenever delegation is suboptimal (case i), the frequency of supervisory interventions decreases. This theoretical prediction is starkly inconsistent with our empirical evidence, which demonstrates an *increase* in interventions following the reform. We can therefore deduce that the supervisory system operates within the third parameter range: the local bias is sufficiently small, informative communication exists, and delegation is both optimal and intervention-maximizing.¹⁷

3.6 Two message equilibrium

In this section we derive several predictions of the model that can be empirically examined. We contrast them with the predictions of alternative channels, unrelated to supervisory communication, for why decentralization may affect interventions.

We restrict the analysis to the empirically relevant parameter range. First, we assume that we are in the third range of Proposition 2 ($b < \bar{b}$). Second, the analysis in Section 5.2 shows that we are in the case of $n^* = 2$ (that is, $1 + \sqrt{\frac{1+\delta}{2b}} < 3$). Furthermore we set $\bar{\delta} = 0$ as $\bar{\delta}$ does not affect intervention frequencies in the third range.

Assumption 1. $\frac{1}{8} < b < \frac{1}{2}$ and $\bar{\delta} = 0$.

From Proposition 1 we have that the threshold for sending the high message is

$$a_1 = \frac{4}{3}b + \frac{1}{3}.$$

If θ is below this threshold, the agent will thus send the low message and the principal sets a fine of $F_1 = 0$. When θ is above the threshold, the agent sends a high message and the principal will set a fine equal to its expected preferred fine in the high region: $F_2 = \frac{1+a_1}{2}$. Given that $a_1 > b$, it follows that there are hence strictly less interventions than under decentralization (in Section 5.2 we quantify this effect based on our empirical estimates).

¹⁷In practice, the likelihood of an intervention also depends on risk-taking by banks. The evidence in Section 5.6 suggests that risk-taking actually declines following the reform. This implies that the estimated positive effect of the reform on interventions is biased downwards; the true increase in supervisory effectiveness is likely even stronger, further reinforcing that the system resides in this third parameter regime.

We can confirm that this is indeed an equilibrium. First, at the threshold a_1 , the agent is exactly indifferent to sending a high or low message. Second, we have that $F_1 = 0$ is optimal following m_1 since $a_1 < 1$. Moreover, the assumption $b < \frac{1}{2}$ guarantees that $a_1 < 1$.

3.6.1 Average fines and their dispersion

Under delegation, the average fine in a supervisory intervention ($E(F|F > 0)$) is lower (the average fine under centralization is $\frac{1+a_1}{2}$, whereas the one under decentralization is $\frac{1-b}{2}$). There are two reasons for this. First, the threshold for intervention declines (since $b < a_1$). That is, there are now also interventions in banks with lower θ , which warrant lower fines. Second, under delegation the local supervisor sets the fines, and his preferred fine is lower.

Whereas average fines are lower under decentralization, the (normalized) deviation of this fine ($\frac{E(|F-E(F|F>0)|)}{E(F|F>0)}|F > 0)$) is higher. Intuitively, this is because under centralization the agent partitions information into intervals, so information about the value of θ within each interval is lost. In the two message equilibrium fine variability is even zero as there is only a single positive fine (F_2).

A noteworthy implication of our model is that the total bias in interventions (combining the extensive and intensive margin) decreases. This bias can be (inversely) represented by the total disbursed fines ($E(F)$). Total fines under decentralization are $E[F] = \frac{1-b}{2} \frac{1-b}{2}$ (the product of the likelihood of intervention $E[F > 0] = \frac{1-b}{2}$ and the average fine issued $E[F|F > 0] = \frac{1-b}{2}$) and total fines under centralization are $E[F] = \frac{1-a_1}{2} \frac{1+a_1}{2}$ (the likelihood of intervention and the average fine are $E[F > 0] = \frac{1-a_1}{2}$ and $F_2 = \frac{1+a_1}{2}$). The latter expression strictly less than $\frac{1-b}{2} \frac{1-b}{2}$ unless we have $b = 0.2$ (in which case there is equality).

3.6.2 Access to local information

Suppose that the principal sometimes also has access to information of the local supervisor. Specifically, assume that with a certain probability she observes θ as well. In such a situation she no longer requires information from the agent under centralization and will simply intervene when $\theta > 0$. Central interventions thus become more frequent (as they otherwise only occur when $\theta > a_1$). It follows that the effect of decentralization on the likelihood of intervention falls (though

our estimates show that they are still positive). The intuition is that when the central supervisor is more informed, the agent's bias will play less a role in centralized supervisory decisions. Thus intervention decisions under centralization are less distorted, lowering the impact of delegation.

3.6.3 Local uncertainty

The benefit of having access to local information is higher when local conditions are less predictable, that is, when they are more uncertain (Dessein, 2002). To understand the effect of uncertainty in our setting, let $0 < \bar{\delta} < b$ instead of $\bar{\delta} = 0$ and assume that with certain probability $\theta = 0$ (instead of being distributed on $[-1, 1]$). In such a situation, the decentralized supervisor would never intervene, whereas the central supervisor would intervene with 50% probability (when $\delta = \bar{\delta}$). Thus, removing uncertainty reverses the effect of decentralization on interventions. Given that the overall effect of decentralization is positive in our data, higher uncertainty increases the effect of decentralization on interventions.

3.6.4 Competing channel I: Higher risk of local banks

Around the time of the reform, there may have been a potentially confounding effect that has increased the risks of local banks (but not national banks). This may potentially explain why supervisory interventions have increased following the reform, unrelated to supervisory communication.

To examine, assume that the informational frictions of our model are absent. That is, the central supervisor also observes θ (either because she directly observes it, or because the local supervisor relays it truthfully to her). Consider now, that for exogenous reasons, the risks of banks increases after the reform. Specifically, assume that the state θ is now uniformly distributed on $[-1 + r, 1 + r]$ ($r \in (0, 1)$) instead of $[-1, 1]$.

Prior to the reform, the fee under centralization is set according to $F = \max\{0, \theta\}$. The resulting likelihood of intervention is $\frac{1}{2}$ and the average fine in an intervention is $\frac{1}{2}$. Following the reform, the fee is set according to $F = \max\{0, \theta - b\}$. The likelihood of intervention is $\frac{1+r-b}{2}$ and the average fine is $\frac{1+r-b}{2}$. We can see that interventions can indeed increase when the risk effect is sufficiently

large ($r > b$). However, in this case, fees should also increase, inconsistent with the evidence. The intuition is simple: higher interventions require local banks to become sufficiently more risky, but in this case, this would also imply higher fees.

3.6.5 Competing channel II: Higher stringency of local supervisors

The reform may also have increased the stringency of the local supervisor, specifically towards local banks. Again, this may cause more interventions into local banks, unrelated to an informational channel.

To investigate, we now assume that the reform lowers the bias of the local supervisor by Δb ($\Delta b > 0$). Outcomes prior to the reform, that is, outcomes under centralization, are unchanged and identical to the previous subsection (the likelihood of intervention and the average fine are both $\frac{1}{2}$). Following the reform, the fee will be set according to $F = \max\{0, \theta - b + \Delta b\}$. The likelihood of intervention is $\frac{1-b+\Delta b}{2}$ and the average fine is $\frac{1-b+\Delta b}{2}$. We see that interventions can indeed increase when Δb is sufficiently large ($\Delta b > b$), but in this case the average fee should increase as well, and not decrease.

3.6.6 Competing channel III: Increased supervisory capacity

Suppose that following the reform local supervisors are endowed with additional resources (such as more manpower). This may again affect interventions irrespective of supervisory communication.

We model increased resources as resulting in higher precision of a signal about the true state of the bank θ . Assume (as in Section 3.6.4) that informational frictions are absent (that is, the central supervisor also observes the signal received by the local supervisor). Suppose that with certain probability this signal takes the true value θ and with remaining probability the signal is only noise. In the first case, the central supervisor observes the true state of the bank. As discussed in Section 3.6.4, this results in an intervention likelihood of $\frac{1}{2}$ and an average fine of $\frac{1}{2}$. In the second case, she is essentially indifferent to interventions. This can be interpreted as intervening with equal likelihood (probability $\frac{1}{2}$) and imposing a fine $F = 0$ upon intervention.¹⁸

¹⁸In the general model of Section 3, this intervention behavior becomes the unique outcome when letting $\bar{\delta} \rightarrow 0$.

It follows that higher supervisory capacity (interpreted as a high likelihood of observing the true state of the world) results in higher fines upon intervention. The intuition is when the supervisor has low information (low capacity), she is uninformed about the true risk of the bank. Since the priors are that average misconduct is low in each year, this results in low fines when the supervisor decides to intervene.

4 Data and Empirical Design

4.1 Sample composition and data sources

Supervisory interventions and resulting penalties are disclosed on the CBRC/CBIRC websites. The information disclosed contains the date, the affected institution or individual, the responsible supervisory authority, the reasons underlying the intervention, the specific laws or regulations violated, and the penalty (or penalties) imposed.

There are five types of penalties. The by far most common one is a fine, which is a monetary penalty imposed on a bank (like the civil money penalty in the US context). Besides fines there are warnings (a formal letter issued by a supervisor stating non-compliance with laws or regulations), a disqualification barring bank managers from holding senior positions in the banking sector, a prohibition barring bank staff from working in the banking sector and a revocation of the license to operate the branch. Figure IA2 in the Internet Appendix provides an example of a penalty as disclosed on the CBRC/CBIRC website.¹⁹

We construct the sample of supervisory interventions using textual analysis of penalty announcements. Our sample includes 12,078 penalty events during the period between 2010 and 2020. Table 1 shows statistics on penalties. Panel A reports the frequency of penalties by types, showing that fines and warnings are the most issued type of penalties (63% and 32%). Panel B breaks down penalties based on their underlying reasons, showing that in more than 50% of the cases the main reason is loan related (note that a bank may receive a penalty for several reasons). Panel C breaks

¹⁹In 2015, the Yichang branch of Bank of Hubei was punished for inaccurately classifying loans based on their risk level and withholding a certain percentage of the loan as a deposit when the loan was granted. Consequently, the branch received a fine of RMB 400,000 from the local supervisory office, Yichang Bureau (Figure IA2).

down penalties based on their recipient (an individual or a bank), showing that banks are the most frequent recipients.

Penalties are aggregated at the prefecture branch level (containing both penalties to the branch itself and to lower-level offices) as the prefecture is the perimeter of the local supervisors. The final sample covers 5,429 (prefecture) branches of 1,074 banks in 342 prefectures for ten years surrounding the 2015 reform. The 1,074 banks consist of 1,056 local banks (127 prefecture commercial banks and 929 rural commercial banks²⁰) and 18 national banks (6 large state-owned banks and 12 joint-stock banks). Overall, our sample represents about 90% of the assets of the Chinese banking sector.

Table IA1 in the Internet Appendix presents additional information about local and national banks at the branch, bank and aggregate levels. One key takeaway from this table is that local and national banks are fairly comparable at the branch-level. Branches of both types of banks are similar in size and profitability and operate in prefectures with comparable market concentration, credit availability, and distance to Beijing. The main difference is that branches of national banks have a modestly higher market share than those of local banks (6.56% versus 5.03%).

We supplement the penalty data with bank-level, firm-level, prefecture-level, and loan-level data from the Chinese Research Data Services (CNRDS) and the China Stock Market and Accounting Research (CSMAR). Sample summary statistics are presented in Table 2. All variable definitions are given in Appendix B.

4.2 Empirical specification

We use a difference-in-differences design to examine the impact of the decentralization reform of 2015 on supervisory interventions. We estimate the following specification at the bank-prefecture-year level:

$$Penalty_{ijt} = \alpha_i + \alpha_j + \alpha_t + \beta Local\ bank_i \times Post_t + \varepsilon_{ijt}, \quad (2)$$

where the subscript i denotes a specific bank i , j the prefecture of the bank branch, and t the year. The dependent variable, $Penalty_{ijt}$, is a one of our measures of penalties imposed on branch belonging to bank i in prefecture j in year t . We use two main variables capturing supervisory

²⁰We exclude all other banks such as rural credit cooperatives and village banks. We also exclude foreign banks.

interventions. The first variable is a dummy taking the value of 1 if a bank branch receives at least one penalty in year i , and 0 otherwise. The second variable is the log of 1 plus the number of penalties received by a bank branch in year i . In further tests, we also create similarly constructed variables for each type and each recipient of penalties. *Local bank _{i}* is a dummy taking the value of 1 for branches of local banks i (treated group), and 0 for branches of national banks i (control group). *Post _{t}* is a dummy taking the value of 1 from 2015 onwards, and 0 otherwise. The bank fixed effects (α_i) control for differences across banks, while the prefecture fixed effects (α_j) control for differences in local conditions. Since each prefecture has one supervisory office, the prefecture fixed effects also account for any (time-invariant) differences in local supervisory stringencies. We also include year fixed effects (α_t) to control for any macro movements. In some specifications, we further include prefecture $\times$ year fixed effects, which control for time-varying prefecture-level heterogeneity. For instance, there may be a turnover of supervisory officers, resulting in a change in local supervisory leniency. In other specifications, we also include bank $\times$ prefecture fixed effects (equivalent to branch fixed effects), which control for heterogeneity across branches of the same bank. Our main specification does not include endogenous bank-level controls to avoid the “bad control” problem (see, e.g., [Gormley and Matsa, 2014](#); [Angrist and Pischke, 2009](#)). ε_{ijt} is the error term. The coefficient of interest is β , which measures the effect of the 2015 decentralization reform on treated branches. We expect β to be positive if local banks are subject to tighter supervision following the reform. Throughout, we report robust standard errors clustered at the level of the prefecture where the branch is located.²¹

5 Empirical Results

5.1 Penalties

Baseline results We begin our analysis by examining the effect of the 2015 decentralization reform on supervisory interventions. Table 3 presents the results from estimating equation 2. The

²¹We experiment with alternative clustering levels—in particular, clustering by bank and double clustering by prefecture and year. Clustering by prefecture used in the reported results produces the most conservative standard errors.

first four columns focus on the likelihood of receiving a penalty. In column 1, we run the regression without any fixed effects. In column 2, we add separate bank, prefecture, and year fixed effects, implying that only the interaction between $Local\ bank_i \times Post_t$ is estimated. In column 3, we include prefecture $\times$ year fixed effects to absorb any time-varying local shocks, meaning that we compare branches of national versus local banks in the same prefecture of the same year. In column 4, we include bank $\times$ prefecture fixed effects and year fixed effects, which controls for differences across branches.

In columns 1 to 4, the estimated coefficients on the interaction term, which are statistically significant at the 1% level, show that local banks are 6.3 to 8.8 pp more likely to receive a penalty than national banks following the decentralization reform. Given the unconditional probability of receiving a penalty of 11.1% as reported in Table 2, this corresponds to an increase of 57 to 80%.²² In columns 5 to 8, we examine the number of interventions as the outcome and confirm the previous findings. These regressions use the log of one plus the number of penalties to retain observations with zero-valued outcomes (Table IA3 in the Internet Appendix reports results from a Poisson regression model instead).²³ Importantly, these estimated effects are net of any concurrent changes in bank risk-taking induced by the reform. As shown in Section 5.6, bank risk-taking declines following the reform, which would itself be expected to reduce supervisory interventions. Therefore, our estimates likely understate the direct effect of decentralization on supervisory interventions.

Controls and differential trends Table 4 reports the results after additionally controlling for bank- and branch-level characteristics, as well as for characteristic-specific effects of the reform. In the first two columns of Panel A, we control for bank size, the capital adequacy ratio, the Z-score, the net interest margin, the loan-to-deposit ratio, and the loan-to-asset ratio. The inclusion of these controls reduces the sample size by 734 banks, primarily small and unlisted banks that are

²²Table IA2 in the Internet Appendix shows that the results are robust to excluding state-owned banks or penalties issued by the CBRC’s local office in Beijing. In the latter case, the central and local offices are located in the same prefecture, which could blur the distinction between centralized and decentralized supervision.

²³Table IA4 in the Internet Appendix further shows that the results qualitatively hold for each type (Panel A) and each recipient (Panel B) of penalties employing a specification similar to column 4 of Table 3 (Table IA3 contains the corresponding Poisson regressions). We do not conduct regressions for license revocations as there are only three instances of such type of penalties.

not required to publicly disclose their financial statements. In the next two columns, we allow the treatment effect to vary with bank characteristics.²⁴ To do so, we interact each bank characteristic, averaged over the pre-reform period, with $Post_t$.

In Panel B, we control for branch-level characteristics using firm-level data from the National Tax Survey Database (NTSD), which is jointly collected by the State Administration of Taxation of China and the Ministry of Finance of China (SAT-MOF) using a stratified random sampling design (Liu and Mao, 2019). We match branch names and cities to our penalty sample, yielding 3,037 matched branches. We construct branch-level controls for size (log of total assets in RMB) and profitability (return on assets).²⁵

The results in Panels A and B are consistent with our baseline findings in Table 3. However, these specifications should be interpreted with caution because bank and branch characteristics are likely endogenous to supervisory activity (Angrist and Pischke, 2009).

Parallel trends We also examine the dynamics of the effect. Figure 3 shows a version of our baseline specification that interacts the variable $Local\ bank_i$ with the time in years relative to the decentralization reform. Panel A plots the estimated coefficients for the likelihood of a penalty, while Panel B does the same for the number of penalties. The parallel trends assumption holds as there are no visible differences between the treated group and control group prior to the reform. We see a delayed response in interventions to the reform, consistent with a lag between initial suspicion of misconduct and the conclusion of a full supervisory investigation, as well as a general adjustment period to new supervisory procedures. In year 2 following the reform, we observe a permanently higher incidence of penalties. Table IA5 in the Internet Appendix also shows the regressions of the dynamics of the effect of decentralization (omitting the year 2015 as the benchmark). Again, no statistically significant effect exists in the years prior to the reform, and a clear increase appears following it.

²⁴For example, in 2015 China downgraded the loan-to-deposit ratio as a supervisory tool, which may have differentially affected banks with higher loan-to-deposit ratios.

²⁵The quality of the NTSD data varies substantially. For example, reported equity is negative for a large fraction of firms, preventing the construction of reliable equity-based measures. For a detailed description of the NTSD, see Liu and Mao (2019).

Placebo tests A potential concern in difference-in-differences analysis is that serial correlation may bias standard errors, in turn leading to over-rejection of the null hypothesis of no effect (Bertrand et al., 2004). We address this concern by performing a permutation test following Chetty et al. (2009) and Ohn (2018), among others. We start the procedure by randomly selecting a placebo implementation year between 2010 and 2020 for each permutation. Then, we randomly designate 18 banks from the entire sample and assign them (and their branches) the status of national banks, while treating the remaining banks and their branches as local banks. The baseline regression, reported in column 4 of Table 3, is then re-estimated for each of our two dependent variables using the placebo treatment. Point estimates are recorded, and the procedure is repeated another 499 times to produce the plots in Figure 4. Both panels of the figure display the empirical distribution of placebo effects for both dependent variables. Reassuringly, the estimated coefficients are normally distributed around 0 and are far away from the actual estimated effects.

5.2 Interventions distortions and efficiency gains

We can use the estimated coefficients from Table 3 to gauge how much the bias distorts interventions and how informative communication between local and central supervisors is. For this we generalize the model in Section 3 by assuming that only with probability p ($\in (0, 1)$) each year a situation materializes in which an intervention is possibly desirable (for instance, because a negative event at a bank materializes). In such a situation, the model described in Section 3 begins, and with remaining probability no interventions take place.

Denote the probability of an intervention under local and central supervision with p_L and p_C , respectively. From column 4 of Table 3, we have that $p_L - p_C = 8.8\%$. The overall likelihood of intervention in the sample is 11.1% and the fraction of observations that are falling under local supervision throughout the sample period is 23.4%. From this we can calculate $p_L = 17.7\%$ and $p_C = 9.0\%$. We have for the likelihood of intervention that $p_L = p \frac{1-b}{2}$ and $p_C = p \frac{1-a_1}{2}$. From the fact that interventions increase (that is, $p_L > p_C$), we know that we are in the parameter range of part (iii) of Proposition 2. We can then use the equations for n^* and a_1 in Section 3.4 to obtain numerically the (unique) solution $b = 0.196$, $a_1 = 0.595$, $n^* = 2$ and $p = 44.1\%$.

The implied values suggest that the preferences of the local supervisor are considerably more lenient than of the central supervisor: In about 19.6% of cases where the central supervisor would want to intervene, the local supervisor does not want to intervene. Interventions are much more biased under central supervision: In about 59.5% of cases where the central supervisor would want to intervene, she ultimately does not intervene. As explained in Section 3.5, this is because strategic communication by the local supervisor reduces the informativeness of information that arrives at the central level.

The inefficiency of communication is also reflected in a low n^* , the number of messages that are sent is equal to the minimum level ($n^* = 2$). That is, the local supervisor either sends a low message indicating that bank risk is low (following which the central supervisor does not intervene), or he sends a high message, following which there is intervention. Under this communication, significant information about the true health of the banks is withheld as within each risk classification (*low*, *high*) there is no longer information about the true underlying risk (the realization of θ). This means that supervisory decisions under central supervision do not well reflect bank conditions.

We can also calculate the model-implied welfare gain due to the reform, which is $\frac{-U_C + U_L}{-U_C} = 18.8\%$. That is, the reform eliminates 18.8% of the losses from inefficient interventions under centralized supervision,²⁶ arising in particular because now also more intermediate forms of misconduct are caught.²⁷ Coupled with the fact that our analysis has shown that there are large distortions under centralized supervision, this suggests that the reform has material welfare benefits.

5.3 Average fines and their dispersion

A prediction of our model is that the severity of interventions declines following decentralization, even though interventions become more likely (see Section 3.6.1).

Our data include information on the size of fines, allowing us to test this prediction. Table 5

²⁶We can arrive at an alternative parameterization of the model using the estimated change in the fine severity instead of the change in the intervention probabilities. Using the average of the linear and log-specification estimates (columns 1 and 2 in Table 5) and following similar calculations as above we obtain $b = 0.136$, $a_1 = 0.515$, $n^* = 2$ and $p = 41.1\%$ and a welfare gain of 33.6%.

²⁷Efficient interventions have wider effects in terms of controlling risk-taking of banks (see Section 5.6) and in terms of financial stability. To the extent that those are reflected in supervisory objectives, they are already incorporated in the welfare estimates.

reports the results. The sample is restricted to bank-prefecture-year observations with at least one penalty, and the dependent variable is the average fine per bank-year (a bank may receive multiple fines within a year). Because the distribution of fines is highly skewed, we report results using both the fine amount (column 1) and its log (column 2). In both specifications, the interaction term is negative and statistically significant at the 10% and 5% levels, respectively, consistent with the prediction that more minor violations are punished following decentralization.²⁸

A second prediction of the model is that the severity of punishment becomes more variable following decentralization (Section 3.6.1). Under decentralization, supervisors base their decisions on more granular information, allowing fines to better reflect differences in the severity of misconduct. By contrast, under centralization, the information that reaches the central supervisor is coarser (i.e., more partitioned), leading to less differentiated fines. We test this prediction using the dispersion of fines. The dependent variable is the normalized absolute deviation of fines in a bank-prefecture-year, computed using either the fine amount (column 3) or its logarithm (column 4). The interaction term is now positive and statistically significant at the 5% and 10% levels, respectively, consistent with the prediction that decentralization increases the dispersion of fines.

5.4 Information

Access to local information Our model shows that, under centralized decision-making, local information is lost because local supervisors communicate strategically and selectively with the central supervisor. The resulting distortion depends on the central supervisor’s own information about local banks. The model therefore predicts that decentralization is more beneficial when the central supervisor has limited direct access to local information (see Section 3.6.2), as delegation then yields a greater informational gain.

We test this prediction in Table 6. We proxy the informational gain from decentralization by the (log) distance, in kilometers, between the prefecture in which the branch (of a local bank) is

²⁸In Table 5, we restrict the sample to positive-fine observations and include bank $\times$ prefecture fixed effects. Consequently, bank-prefecture units fined in only one year provide no within-unit variation for estimating the treatment effect, meaning that coefficients are identified by repeat-fined branches. This raises the concern that the specification might not capture the new low-severity interventions on the extensive margin. Reassuringly, our results remain unchanged if we instead include separate bank, prefecture, and year fixed effects, rather than the high-dimensional interactions, ensuring these single-intervention marginal cases are retained.

located and Beijing.²⁹ We then examine whether the effect of the reform on penalties is stronger for branches located farther from the central supervisor.

Columns 1 and 2 include year and bank $\times$ prefecture fixed effects. Whether the dependent variable is the likelihood of receiving a penalty or the number of penalties, the coefficient on the triple interaction term is positive and statistically significant at the 1% level. Columns 3 and 4 exploit only within-bank variation arising from differences in branch locations and, hence, in distance to the central supervisor. These specifications include bank $\times$ year and prefecture $\times$ year fixed effects. The coefficient on the triple interaction term remains positive and statistically significant at the 1% level. Overall, the results indicate that decentralization has a stronger effect when the central supervisor is at a greater informational disadvantage, consistent with limited communication under centralized supervision.³⁰

In columns 5 and 6, we examine whether the informational gain increases linearly with distance or is concentrated among branches located very far from Beijing. We divide branches into terciles based on their distance to Beijing (long, intermediate, and short) and interact these indicators with $Local\ bank_i \times Post_t$. The estimated coefficients on the triple interaction terms increase from the short- to the intermediate- and long-distance groups. Notably, the coefficient for the short-distance group is already economically meaningful, highlighting the importance of eliminating even small informational frictions.

Local uncertainty We previously considered a proxy for the amount of information lost under centralized decision-making. Here, we instead consider the value of local information for supervisory decisions. Our model predicts that local information is more valuable when local conditions are more heterogeneous and therefore harder for the central supervisor to predict (see Section 3.6.3). In such environments, decentralization should yield larger improvements in the detection of misconduct.

We measure local uncertainty using prefecture-level indicators of economic and political uncer-

²⁹Previous literature has offered US evidence that the distance from the firm location to the banking regulator, the Department of Justice, or the Securities and Exchange Commission, affects the costs of monitoring and information acquisition (see, e.g., Kedia and Rajgopal, 2011; Wilson and Veuger, 2017; Ganduri, 2019; Gopalan et al., 2021; Ha et al., 2023; Lim et al., 2023).

³⁰The results are robust to using travel time between the branch and the central supervisor instead of geographic distance in kilometers.

tainty. We proxy economic uncertainty by the standard deviation (SD) and coefficient of variation (CV) of annual GDP growth over the preceding three years (from year $(t - 3)$ to year $(t - 1)$). We then classify a prefecture as having high economic uncertainty if its SD or CV exceeds the sample median in a given year. Accordingly, we construct a dummy taking the value of 1 if a prefecture’s SD or CV is above the annual sample median, and 0 otherwise. In addition, we proxy political uncertainty using disruptions in the leadership of prefecture-level governments. We manually collect the names, tenures, and exact appointment and termination dates of the top Party secretary and mayor for each prefecture during our sample period. We construct two measures of political uncertainty. First, we define a dummy taking the value of 1 if a leadership position remains vacant for more than six months within a year, and 0 otherwise. Second, exploiting China’s five-year norm for local leadership terms, we define another dummy taking the value of 1 if a top leader is replaced within three years of appointment, and 0 otherwise.

Table 7 reports the results for local economic uncertainty in Panel A and political uncertainty in Panel B. Focusing first on economic uncertainty, we find evidence consistent with the model’s prediction. Across all specifications, the coefficient on the triple interaction term is positive and statistically significant, indicating that local supervisors become tighter following the reform when local economic uncertainty is high. The results are stronger when uncertainty is measured using the CV rather than the SD and remain qualitatively unchanged across the two dependent variables. Turning to political uncertainty, we again find evidence consistent with the model’s prediction. The coefficients on the triple interaction term remain positive and statistically significant across both measures of political uncertainty and both dependent variables, indicating that local supervisors likewise become tighter following the reform when political uncertainty is high.

5.5 Competing channels

Our results thus far point to more effective use of local information under decentralization, resulting in improved supervision. We next examine whether the evidence is consistent with alternative explanations.

Risk-taking A higher frequency of supervisory interventions could reflect increased risk-taking by banks rather than changes in supervisory behavior. This explanation would require local banks to have increased their risk-taking relative to national banks around the time of the reform. For example, the de-emphasis of the loan-to-deposit ratio as a supervisory tool in 2015 may have affected local banks differently if they expanded lending more aggressively.

Our evidence is inconsistent with this explanation for several reasons. First, as discussed in Section 3.6.4, higher bank risk should result in larger fines when interventions occur, whereas we find that average fines decline following the reform. Second, Table 4 shows that our results are robust to controlling for bank- and branch-level risk characteristics, as well as allowing the effects of the reform to vary with such characteristics, including the loan-to-deposit ratio. Finally, as shown in Section 5.6, banks become more conservative in their lending following the reform, which is inconsistent with an increase in underlying risk-taking.

Supervisory stringency A higher frequency of supervisory interventions may also reflect an increase in supervisory stringency rather than improved use of local information (Granja and Leuz, 2024). This explanation would require local supervisors to have become stricter toward local banks following the reform (that is, in our notation, a reduction in b for interventions involving local banks). For example, the reform may have increased the accountability of local supervisors, thereby reducing their bias against intervention. However, as explained in Section 3.6.5, this mechanism predicts not only more frequent interventions but also higher fines, whereas we find that average fines decline following the reform.

Supervisory capacity The ability to collect information about misconduct depends on supervisory capacity (Kandrac and Schlusche, 2021; Eisenbach et al., 2022). Following the reform, local supervisors may have hired additional staff, improving their ability to detect misconduct and thereby increasing supervisory interventions.

To account for changes in supervisory capacity, we manually collect data from the CBRC/CBIRC website on new hires at the provincial level. Columns 1 and 2 of Table 8 augment the baseline specification with the log of the number of new hires. In columns 3 and 4, we restrict the sample to

the period before 2018 because, following the merger of the CBRC and the CIRC (see Section 2.2), it is no longer possible to identify hires specifically related to bank supervision. Our main results remain robust across all specifications.

Moreover, greater supervisory capacity is inconsistent with our finding that average fines decline following the reform. As explained in Section 3.6.6, limited information about misconduct leads supervisors to impose smaller fines because they are less certain about the severity of violations. Better information resulting from higher supervisory capacity should therefore increase, rather than decrease, the average fine.

5.6 Lending

Loan-level analysis Better detection of misconduct should reduce banks’ incentives to take on risk. We test this prediction using individual lending decisions at the branch level. Focusing on individual loans allows us to control for a rich set of borrower characteristics, including borrower fixed effects.

Specifically, we collect loan announcements for all listed firms in China over our sample period. Using textual analysis, we extract information on borrower identity, loan origination dates, loan amounts, loan spreads, and the originating bank branch. The initial sample consists of 16,184 loans issued to 1,678 firms by 327 banks. We obtain borrower financial information from CNRDS, including firm size, leverage, tangibility, cash holdings, and return on assets. Merging these data with our branch-level information yields a final sample of 13,358 loans.

We measure lending conservatism using loan spreads (interest rates) and loan quantities (log of loan amounts). Conditional on borrower characteristics, a more conservative bank branch is expected to charge higher interest rates to compensate for credit risk and to issue smaller loans, thereby taking on less risk. Variable definitions and summary statistics are reported in Appendix B and Table 2, respectively.

Table 9 reports the results. We first examine loan spreads. In column 1, the specification includes lagged borrower characteristics together with year, bank, and borrower fixed effects. Column 2 additionally includes prefecture fixed effects. In both specifications, the number of observations

is relatively small because loan spread information is often unavailable. The estimated coefficients on the interaction term are positive and statistically significant at the 5% level, indicating that local banks charge higher loan spreads following the 2015 reform. The estimated effects amount to 31.3-31.5 basis points, corresponding to approximately 29% of the standard deviation of loan spreads. These findings are consistent with local banks adopting more conservative lending policies following the reform by requiring higher compensation for lending to otherwise identical borrowers. Most borrower characteristics are statistically insignificant, suggesting that borrower risk is largely time-invariant and therefore well captured by the borrower fixed effects.

In columns 3 and 4, we focus on loan quantities. The estimated coefficients on the interaction term are negative and statistically significant at the 1% and 5% levels, respectively, indicating that local banks reduce the amount they lend following the 2015 reform. Economically, the estimated effects correspond to a 28-32% reduction in loan size, also consistent with increased conservatism resulting from decentralization.³¹

Prefecture-level analysis In Table [IA6](#) in the Internet Appendix, we investigate whether these changes translate into aggregate effects at the prefecture level. Specifically, we relate prefecture-level loan supply (measured as the ratio of credit to GDP) to the interaction between the post-reform period and the share of local banks in the prefecture. To mitigate endogeneity in the share of local banks, we also estimate an instrumental variable (IV) specification using the predetermined share of local banks in 2010 as an instrument, following [Gilje et al. \(2016\)](#). Across both OLS and IV specifications, we find that prefectures with a higher share of local banks experience a larger decline in loan supply following the reform. These findings suggest that the increase in supervisory stringency induced by decentralization has economically meaningful aggregate effects, despite directly affecting only *local* bank branches.

³¹The number of listed firms increased substantially during our sample period, raising the possibility that the composition of borrowers differs before and after the reform. To address this concern, we re-estimate the specifications in Table [9](#) using only firms that were already listed before the reform and obtain similar results.

6 Conclusion

This paper studies the effects of decentralizing bank supervision in China’s banking sector. Following the 2015 transfer of supervisory authority over local banks from national to prefecture-level supervisors, the probability of supervisory intervention increased by 57–80%. Additional evidence suggests that decentralization improved supervision through more effective use of local information. Combined with our model of supervisory communication under local bias, our empirical estimates indicate that the reform lies in the welfare-enhancing region of delegation.

Policy discussions, particularly in the Eurozone, have largely focused on centralized supervision relative to independent national supervisors. Our analysis highlights the role of decentralization within hierarchical supervisory organizations. When local supervisors report to a central authority, strategic communication distorts the transmission of local information. Delegating intervention decisions mitigates this distortion while preserving centralized objectives. Our results thus challenge the view that decentralization leads to more biased and less effective supervision. In hierarchical organizations, decentralization can instead improve supervisory efficiency and welfare. This insight extends beyond bank supervision to other hierarchical organizations in which lower-level managers hold private information (e.g., about misconduct or investment opportunities) but have incentives to withhold or distort it.

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A Proofs

A.1 Proof of Proposition 1

Proof. Fix δ . We ignore knife-edge equalities, which can be handled by the usual tie-breaking conventions and do not affect the characterization. Any informative cheap-talk equilibrium can be represented by an ordered partition $-1 = a_0 < a_1 < \dots < a_n = 1$ and messages $M = \{m_1, \dots, m_n\}$ such that the agent sends m_k whenever $\theta \in (a_{k-1}, a_k]$.

The principal chooses F_k to minimize the expected absolute distance from her ideal fine, conditional on receiving message m_k . Because the loss function is absolute value, the optimal fine is a median of the posterior distribution of $F_P^*(\theta)$. Since θ is uniformly distributed on $(a_{k-1}, a_k]$ after message m_k , and since $F_P^*(\theta) = \max\{0, \theta + \delta\}$ is weakly increasing in θ , this implies

$$F_k = \max \left\{ 0, \frac{a_{k-1} + a_k}{2} + \delta \right\}.$$

At every interior cutoff a_k , the agent must be indifferent between sending m_k and m_{k+1} . This gives

$$2(a_k + \delta) = F_k + F_{k+1} + 2b. \tag{3}$$

If two adjacent messages induce the same action, then they are equivalent to a single message. We therefore impose without loss of generality that $F_k > 0$ for all $k > 1$.

Equilibria with $F_1 = 0$. First consider equilibria in which the lowest message induces no fine. By the action formula above, this requires

$$\frac{a_1 - 1}{2} + \delta < 0,$$

or equivalently $a_1 + \delta < 1 - \delta$. For $k > 1$, the principal's action is interior, so $F_k = (a_{k-1} + a_k)/2 + \delta$.

Using (3), the first cutoff satisfies

$$a_1 + \delta = \frac{1}{2}(a_2 - a_1) + 2b,$$

and the remaining cutoffs satisfy

$$a_k - a_{k-1} = a_{k+1} - a_k + 4b, \quad k = 2, \dots, n-1.$$

Let $x \equiv a_n - a_{n-1}$ be the length of the highest interval. Then, for $k = 2, \dots, n$,

$$a_k - a_{k-1} = x + 4(n-k)b.$$

The first cutoff is therefore pinned down by

$$\begin{aligned} a_1 + \delta &= 2b + \frac{1}{2}(a_2 - a_1) \\ &= 2(n-1)b + \frac{x}{2}. \end{aligned}$$

Using $a_n = 1$, we obtain

$$\begin{aligned} 1 + \delta &= a_1 + \delta + \sum_{k=2}^n (a_k - a_{k-1}) \\ &= 2b(n-1)^2 + \left(n - \frac{1}{2}\right)x. \end{aligned}$$

Thus

$$x = \frac{(1 + \delta) - 2b(n-1)^2}{n - 1/2}.$$

A partition with n messages exists if and only if $x > 0$, or

$$2b(n-1)^2 < 1 + \delta.$$

Hence the most informative equilibrium has

$$n^* = \left\lfloor 1 + \sqrt{\frac{1+\delta}{2b}} \right\rfloor,$$

and informative communication is feasible if and only if $n^* > 1$, which is equivalent to

$$b < \frac{1}{2} + \frac{\delta}{2}.$$

Substituting the value of x into the expression for $a_1 + \delta$ gives

$$a_1 + \delta = b(n^* - 1) \left(\frac{2n^*}{2n^* - 1} \right) + \frac{1 + \delta}{2n^* - 1}.$$

The remaining interval lengths are then determined recursively by

$$a_2 - a_1 = 2(a_1 + \delta) - 4b,$$

$$a_{k+1} - a_k = a_k - a_{k-1} - 4b, \quad k > 1.$$

Finally, recall that the condition that $F_1 = 0$ equals $a_1 + \delta < 1 - \delta$. Substituting a_1 , we have

$$b < 1/n^* - \delta/(n^* - 1).$$

One can verify that for $\delta \in (0, 1/3)$, the relevant condition is the one where, at the cut-off for b , $n^* = 2$. For $\delta \in (1/3, 1/2)$, the relevant condition has $n^* = 3$ at the cut-off. Thus, the condition for $F_1 = 0$ to be optimal following m_1 is given by

$$b < \bar{b} \equiv \max\{1/2 - \delta, 1/3 - \delta/2\}$$

We conclude that an informative equilibrium with a no-intervention message exists whenever

$$b < \min \left\{ \frac{1}{2} + \frac{\delta}{2}, \bar{b} \right\}.$$

Equilibria with $F_1 > 0$. Now consider equilibria in which the principal imposes a strictly positive fine after every message. Then all actions are interior, and (3) implies

$$a_k - a_{k-1} = a_{k+1} - a_k + 4b, \quad k = 1, \dots, n-1.$$

Letting $x = a_n - a_{n-1}$, we have

$$a_k - a_{k-1} = x + 4(n-k)b, \quad k = 1, \dots, n.$$

Since the intervals sum to two,

$$\begin{aligned} 2 &= \sum_{k=1}^n (a_k - a_{k-1}) \\ &= nx + 2bn(n-1). \end{aligned}$$

Thus a positive-action partition with n messages exists if and only if

$$2bn(n-1) < 2.$$

The largest feasible number of messages $n^\#$ is therefore

$$n^\# = \left\lfloor \frac{1}{2} + \frac{1}{2} \sqrt{1 + \frac{4}{b}} \right\rfloor,$$

and informative communication of this form is feasible if and only if $b < 1/2$.

For the first action to be strictly positive, we need

$$\frac{a_1 - 1}{2} + \delta > 0,$$

Since in equilibrium, $a_1 + 1 = x + 4(n-1)b = 2/n + 2(n-1)b$, this is equivalent to

$$b > \frac{1}{n^\#} - \frac{\delta}{n^\# - 1}$$

where $n^\#$ is a function of b . One can show that, for $\delta < 1/3$, the relevant inequality has $n^\# = 2$ and is thus $b > 1/2 - \delta$. When $\delta \in (1/3, 1/2)$, the relevant condition has $n^\# = 3$ and is thus $b > 1/3 - \bar{\delta}/2$. Note that this is the same threshold $\bar{b}$ we obtained as a condition for $F_1 = 0$. Therefore, when $\delta > 0$ and $b \in (\bar{b}, 1/2)$, informative communication may be feasible, but all messages induce strictly positive fines.

Combining the two cases proves the proposition. If $\delta < 0$, the binding condition for informative communication is $b < (1/2) + (\delta/2)$; when this fails, communication is uninformative and the principal chooses no fine because the unconditional median of $\theta + \delta$ is negative. If $\delta > 0$, the principal always issues a positive fine once $b > \bar{b}$. Communication may still be informative for $b < 1/2$, but it cannot affect the extensive margin; for $b > 1/2$, no informative communication is feasible and the principal still issues a fine because the unconditional median of $\theta + \delta$ is positive. $\square$

A.2 Proof of Proposition 2

Proof. 1) Frequency of Intervention: Part (i) and (ii): From Proposition 1, whenever $b > \bar{b}$, there is always intervention under centralization if $\delta > 0$ (which occurs with 50% probability); there is intervention under delegation if and only if $\theta > b - \delta$, which occurs in less than 50% of cases. Hence, interventions are more frequent under centralization. Part (iii): Under delegation, there is no intervention when $\theta < b - \delta$. From Proposition 1, under centralization, there is no intervention when $\theta < a_1$, where $a_1 > b - \delta$. Hence, interventions are more frequent under delegation.

2) Optimality of Delegation.

2A) No informative communication under centralization. We first compare delegation with centralization when there is no communication ($n^* = 1$), and show that in the latter case delegation is optimal for $b < 1 - (\sqrt{2} - 1)\bar{\delta}$, and suboptimal for $b > 1 - (\sqrt{2} - 1)\bar{\delta}$.

a) Whenever $b < 1 - \bar{\delta}$, expected pay-offs under delegation given δ equal

$$EU_{D,\delta} = -\frac{1}{2} \int_{-\delta}^{b-\delta} (\theta + \delta) d\theta - \frac{1}{2} \int_{b-\delta}^1 b d\theta = -\frac{1}{2}b(1 + \delta) + \frac{1}{4}b^2$$

Since $\delta = -\bar{\delta}$ with 50% probability and $\delta = \bar{\delta}$ with 50% probability, we obtain

$$EU_D = -\frac{1}{2}b + \frac{1}{4}b^2$$

b) Whenever $b > 1 - \bar{\delta}$, expected pay-offs under delegation given $\delta = \bar{\delta}$ equal

$$EU_{D,\bar{\delta}} = -\frac{1}{2}b(1 + \bar{\delta}) + \frac{1}{4}b^2;$$

expected pay-offs when $\delta = -\bar{\delta}$ equal

$$EU_{D,-\bar{\delta}} = -\frac{1}{2} \int_{\bar{\delta}}^1 (\theta - \bar{\delta}) = -\frac{1}{4}(1 - \bar{\delta})^2.$$

c) Consider now the expected pay-off under centralization and no communication. Assume first that $\delta = -\bar{\delta} < 0$. Since $\delta < 0$, the principal issues no fine under no communication and its expected pay-off equals

$$EU_{C,-\bar{\delta}} = \frac{1}{2} \left(- \int_{\bar{\delta}}^1 (\theta - \bar{\delta}) d\theta \right) = -\frac{1}{4}(1 - \bar{\delta})^2$$

Assume next that $\delta = \bar{\delta} > 0$. Since $\delta > 0$, the principal issues a fine $F = \bar{\delta}$ under no communication, yielding pay-offs equal to

$$\begin{aligned} EU_{C,\bar{\delta}} &= \frac{1}{2} \left(\int_{-1}^{-\bar{\delta}} (-F) d\theta \right) + \frac{1}{2} \left(\int_{-\bar{\delta}}^0 (\theta + \bar{\delta} - F) d\theta \right) - \frac{1}{2} \left(\int_0^1 (\theta + \bar{\delta} - F) d\theta \right) \\ &= \frac{1}{2} \left(\int_{-1}^{-\bar{\delta}} (-\bar{\delta}) d\theta \right) + \frac{1}{2} \left(\int_{-\bar{\delta}}^0 \theta d\theta \right) - \frac{1}{2} \left(\int_0^1 \theta d\theta \right) \\ &= -\frac{1}{4}(1 + 2\bar{\delta} - \bar{\delta}^2) \end{aligned}$$

Since $\delta = -\bar{\delta}$ with 50% probability and $\delta = \bar{\delta}$ with 50% probability, expected pay-offs under centralization (and no communication) equal

$$EU_C = -1/4.$$

It follows that whenever $b < 1 - \bar{\delta}$, delegation is preferred over centralization (without communication) whenever $EU_D > EU_C$ or still:

$$2b - b^2 < 1$$

which is always satisfied for $b < 1$. Whenever $b \in (1 - \bar{\delta}, 1)$, pay-offs under delegation and centralization (no communication) are identical when $\delta = -\bar{\delta}$. In contrast, when $\delta = \bar{\delta}$, $EU_{D,\bar{\delta}} > EU_{C,\bar{\delta}}$ if and only if

$$-\frac{1}{2}b(1 + \bar{\delta}) + \frac{1}{4}b^2 > -\frac{1}{4}(1 + 2\bar{\delta} - \bar{\delta}^2)$$

or still

$$b < 1 - (\sqrt{2} - 1)\bar{\delta}$$

2B) Delegation vs Centralization with informative communication. Assume now that there is informative communication, that is $b < 1/2$ and $n^* > 1$.

Assume first that $\delta = \bar{\delta} > 0$. Then for $b \in (\bar{b}, 1/2)$, there is informative communication but all messages induce strictly positive fines. Any equilibrium with informative communication has then partition elements where $a_k - a_{k-1} = a_{k+1} - a_k + 4b$. This structure of partition elements is the same as in [Dessein \(2002\)](#). We can therefore use the same argument as in [Dessein \(2002\)](#) to show that this implies that whenever $n > 1$, delegation results in a smaller expected absolute loss $|F - F^*|$ than centralization.

Assume next that $b \in (0, \bar{b})$ and either δ is positive or negative. Then under centralization, there is no intervention for $\theta \in (0, a_1)$, whereas under delegation there is no intervention for $\theta \in (0, b - \delta)$. Since $a_1 > b - \delta$ and intervention is optimal for $\theta \in (b - \delta, a_1)$, pay-offs are higher under delegation conditional on $\theta < a_1$. Consider next expected pay-offs conditional on $\theta > a_1$. Whenever $k > 1$, we then have again that $a_k - a_{k-1} = a_{k+1} - a_k + 4b$, so that we can again use the same argument as in [Dessein \(2002\)](#) to show that delegation results in a smaller expected absolute loss $|F - F^*|$ than centralization. $\square$

B Variable Definitions and Data Sources

We provide below detailed definitions for all variables used in our empirical analysis as well as their corresponding data sources.

B.1 Branch-level variables

Local bank: Dummy variable equals to 1 if a bank is not a state-owned bank or joint-stock bank, and 0 otherwise. Source: CBRC/CBIRC.

Penalty dummy: Dummy variable equals to 1 if a bank receives a penalty, and 0 otherwise. A penalty is a punitive measure on a bank enforced by a supervisor as a consequence of significant non-compliance with laws or regulations. Sources: CBRC/CBIRC, authors' calculations.

Number of penalties: Log of 1 plus number of penalties. Sources: CBRC/CBIRC, authors' calculations.

Fine dummy: Dummy variable equals to 1 if a bank receives a fine, and 0 otherwise. A fine includes a monetary penalty imposed on a bank and confiscation of its illegal proceeds. Sources: CBRC/CBIRC, authors' calculations.

Fine amount: Log of 1 plus fine amount (thousand RMB). Sources: CBRC/CBIRC, authors' calculations.

Average fine: Aggregate fine amount divided by the number of fine incidences. Sources: CBRC/CBIRC, authors' calculations.

Fine dispersion: Normalized absolute deviation of the fine amount. Specifically, for each fine on bank i , this variable is calculated as

$$\frac{|Fine_i - \overline{Fine}_{jt}|}{\overline{Fine}_{jt}},$$

where $\overline{Fine}_{jt}$ is the average fine amount across all fines imposed on local or national banks in prefecture j during year t . In other words, the sample of local banks is averaged at the prefecture-year level, separately from the sample of national banks. Sources: CBRC/CBIRC; authors' calculations.

Fine dispersion (log): Absolute deviation of the log fine amount from the mean log fine amount within the same prefecture-year:

$$\left| \log(Fine_i) - \overline{\log(Fine)}_{jt} \right|,$$

where $\overline{\log(Fine)}_{jt}$ is the average log fine amount across all fines imposed on local or national banks in prefecture j during year t . Sources: CBRC/CBIRC; authors' calculations.

Warning dummy: Dummy variable equals to 1 if a bank receives a warning, and 0 otherwise. A warning is a formal notification letter issued by a supervisor, alerting a bank of its non-compliance with laws or regulations. Sources: CBRC/CBIRC, authors' calculations.

Number of warnings: Log of 1 plus number of warnings. Sources: CBRC/CBIRC, authors' calculations.

Disqualification dummy: Dummy variable equals to 1 if a bank receives a penalty that its manager is disqualified and barred from holding positions of senior managers in the banking industry for a specified period or permanently, and 0 otherwise. Sources: CBRC/CBIRC, authors' calculations.

Number of disqualifications: Log of 1 plus number of disqualifications. Sources: CBRC/CBIRC, authors' calculations.

Prohibition dummy: Dummy variable equals to 1 if a bank receives a penalty that its staff is prohibited or banned from working in the banking industry for a specified period or indefinitely, and 0 otherwise. Sources: CBRC/CBIRC, authors' calculations.

Number of prohibitions: Log of 1 plus number of prohibitions. Sources: CBRC/CBIRC, authors' calculations.

Distance: Log of distance in kilometers between the bank branch and Beijing. Sources: Baidu Map, authors' calculations.

Share of offices outside the prefecture: The proportion of the number of bank offices located outside the prefecture where the bank branch operates. Sources: CBRC/CBIRC, authors' calculations.

Branch size: Log of a branch's total assets (million RMB). Source: NTSD.

Branch leverage ratio: Branch's equity over total assets (%). Source: NTSD.

Branch ROA: Branch's net income over total assets (%). Source: NTSD.

B.2 Bank-level variables

Bank size: Log of a bank's total assets (billion RMB). Source: CNRDS.

CAR: Capital adequacy ratio (%). Source: CNRDS.

Z-score: Sum of equity to asset ratio and ROA divided by standard deviation of ROA. A 3-year rolling window is used when calculating the standard deviation of ROA. Sources: CNRDS, authors' calculations.

NIM: Net interest margin (%). Source: CNRDS.

Loan-to-deposit: Gross loans to total deposits (%). Source: CNRDS.

Loan-to-asset: Gross loans to total assets (%). Source: CNRDS.

B.3 Loan-level variables

Loan spread: Loan spread over the benchmark interest rate (basis points). Source: Authors' collection.

Loan amount: Log of loan amount (million RMB). Source: Authors' collection.

B.4 Firm-level variables

Firm size: Log of total assets of a borrower (million RMB). Source: CNRDS.

Firm leverage: Total liabilities to total assets of a borrower (%). Source: CNRDS.

Firm tangibility: Total property, plant, and equipment to total assets of a borrower (%). Source: CNRDS.

Firm cash holdings: Cash holdings to total assets of a borrower (%). Source: CNRDS.

Firm ROA: Return on assets of a borrower (%). Source: CNRDS.

B.5 Prefecture- and province-level variables

Share of local banks: Share of local banks in terms of number of bank offices in a prefecture.

Sources: CBRC/CBIRC, authors' collection.

Credit to GDP: Private credit to GDP of a prefecture (%). Source: CNRDS.

GDP growth: Growth rate of GDP of a prefecture (%). Source: CNRDS.

Fiscal balance: A municipal (prefecture) government's revenue minus its expenditure, divided by its GDP. Source: CNRDS.

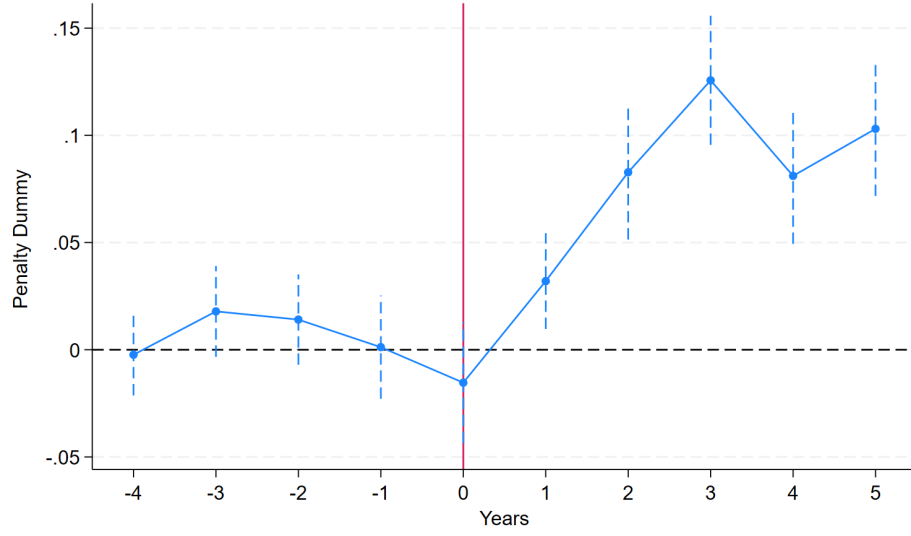
SD of GDP growth: Dummy variable equals to 1 if the SD of GDP growth rates over the past three years is greater than sample median in a given year, and 0 otherwise. Source: CNRDS.

CV of GDP growth: Dummy variable equals to 1 if the CV of GDP growth rates over the past three years is greater than sample median in a given year, and 0 otherwise. Source: CNRDS.

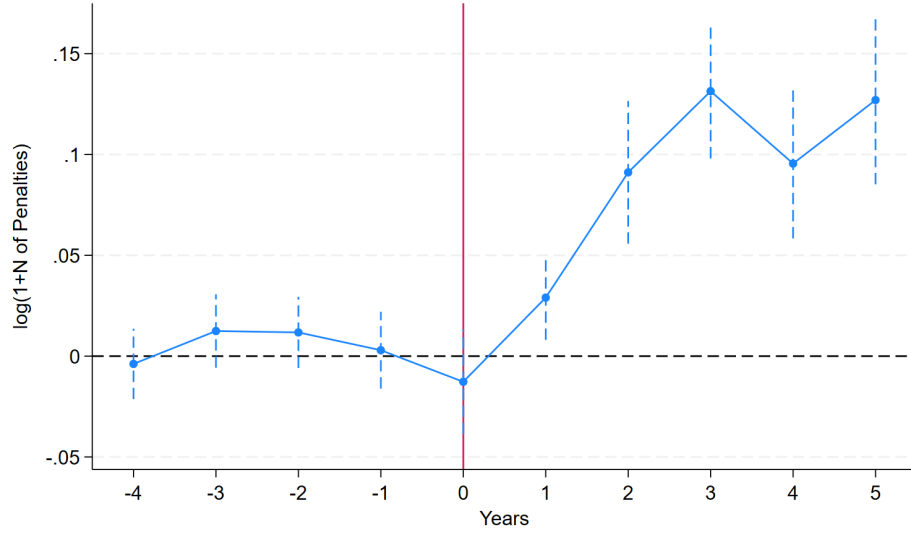
Vacancy>6m: Dummy variable equals to 1 if the local leadership is vacant for over six months, and 0 otherwise. Source: Authors' collection and calculation.

Turnover<3y: Dummy variable equals to 1 if the local leadership is leadership turnover within three years, and 0 otherwise. Source: Authors' collection and calculation.

New hires: Province-level log of number of annual new hires. Source: CBRC/CBIRC.



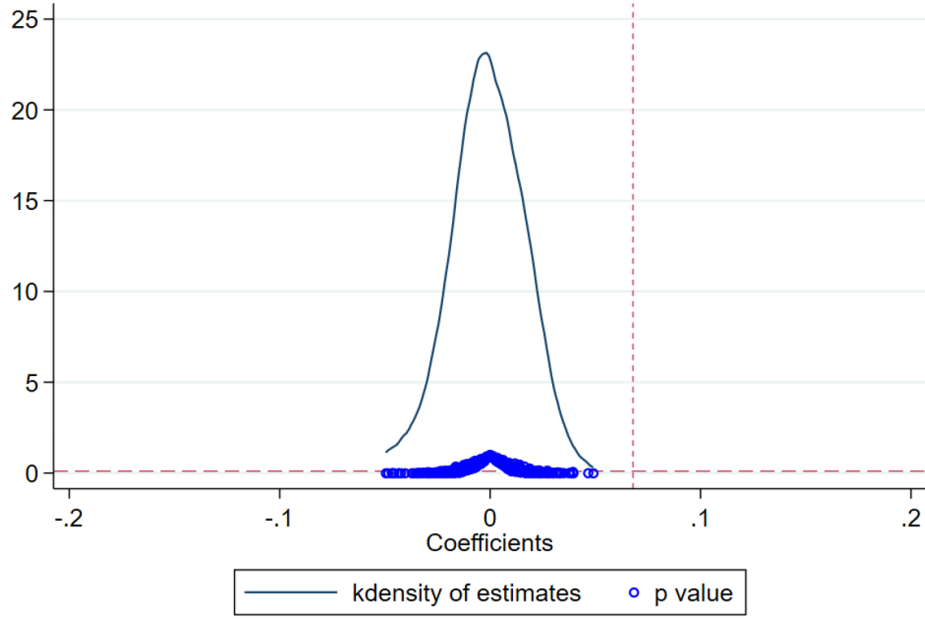
(a) Likelihood of a penalty



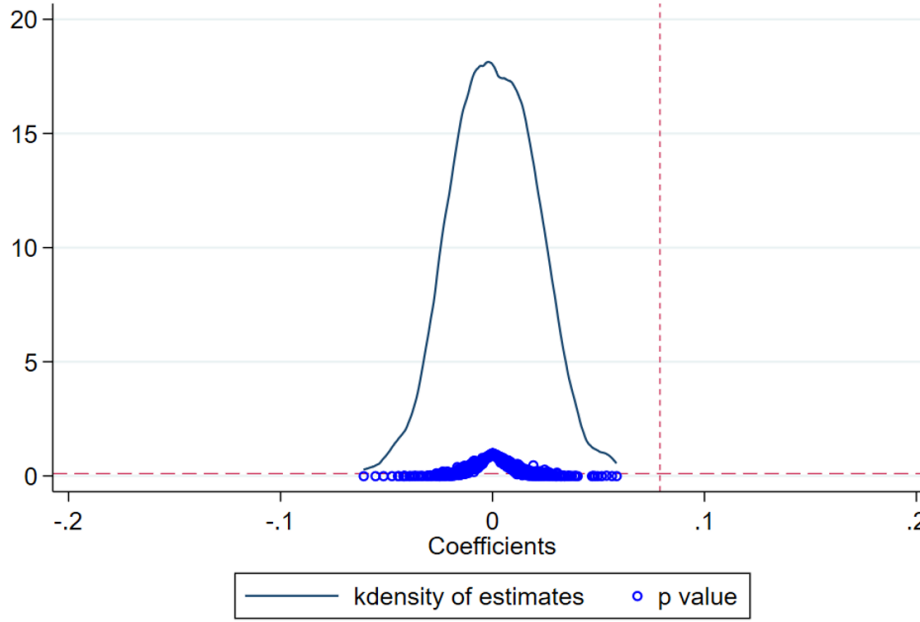
(b) Number of penalties

Figure 3: Parallel trends

The figure presents the event-study estimates from (2), illustrating the dynamic effects of the 2015 decentralization reform. Panel A reports the estimates for the penalty indicator, while Panel B reports the estimates for the log of the number of penalties. Year 0 corresponds to 2015, the year in which the decentralization reform was implemented. The coefficients for the pre-reform years allow an assessment of the parallel trends assumption



(a) Likelihood of a penalty



(b) Number of penalties

Figure 4: Distribution of placebo estimates

The figure presents the empirical distributions of the placebo estimates for the two dependent variables described in Section 5.1. The vertical red dashed line indicates the estimated coefficient from column 4 in Table 3, while the horizontal red dashed line marks the 10% significance level. The solid blue line and blue circles represent the distributions of the estimated coefficients and corresponding p -values, respectively, obtained from 500 placebo tests. Panel A reports the results for the penalty indicator, while Panel B reports the results for the log of the number of penalties.

Table 1: Types, reasons, and recipients of penalties

The table presents summary statistics on penalties imposed on local and national banks. Panel A reports the distribution by penalty type, Panel B by reason for the penalty, and Panel C by recipient. All variables are defined in [Appendix B](#).

Panel A: Type of penalties	N	Local banks	National banks
Fine	8,589	4,050	4,539
Warning	4,341	2,431	1,910
Disqualification	365	192	173
Prohibition	376	194	182
License revocation	3	0	3
Panel B: Reason of penalties	N	Local banks	National banks
Loan-related reasons	6,819	3,688	3,131
Deposit-related reasons	815	271	544
Interbank-related reasons	463	284	179
Acceptance-related reasons	1,534	666	868
Credit Card-related reasons	181	33	148
Guarantee-related reasons	358	154	204
Prudential regulation-related reasons	1,558	771	787
Internal control-related reasons	875	388	487
Governance-related reasons	294	273	21
Panel C: Recipient of penalties	N	Local banks	National banks
Individuals	4,670	2,737	1,933
Banks	7,858	3,513	4,345
Both individuals and banks	450	252	198

Table 2: Summary statistics for all variables

The table presents summary statistics for all variables used in our empirical analysis. Panel A reports them for branch-level variables, Panel B for bank-level variables, Panel C for loan-level variables, and Panel D for prefecture- and province-level variables. All variables are defined in Appendix B.

Panel A: Branch-level variables	N	Mean	SD	P1	Median	P99
Local bank	52,773	0.383	0.486	0	0	1
Penalty dummy	52,773	0.110	0.313	0	0	1
Number of penalties	52,773	0.229	1.050	0	0	4
Fine dummy	52,773	0.105	0.307	0	0	1
Number of fines	52,773	0.163	0.652	0	0	3
Fine amount	52,773	103.318	3,361.840	0	0	1,200
Average fine	3,871	679.844	8704.643	26	300	4513.333
Average fine (log)	3,871	5.696	0.887	3.258	5.704	8.415
Fine dispersion	3,871	0.337	0.320	0	0.260	0.984
Fine dispersion (log)	3,871	0.410	0.441	0	0.266	1.478
Warning dummy	52,773	0.035	0.183	0	0	1
Number of warnings	52,773	0.082	0.670	0	0	2
Disqualification dummy	52,773	0.005	0.068	0	0	0
Number of disqualifications	52,773	0.007	0.117	0	0	0
Prohibition dummy	52,773	0.005	0.071	0	0	0
Number of prohibitions	52,773	0.007	0.118	0	0	0
Distance (km)	52,773	1,152.448	602.336	47.225	1,080.204	2,771.434
Distance (log)	52,773	6.845	0.393	3.876	6.986	7.927
Branch size (log)	18,819	11.788	1.302	9.202	11.756	14.236
Branch ROA (%)	18,823	0.979	1.433	-2.445	0.986	4.195
Panel B: Bank-level variables	N	Mean	SD	P1	Median	P99
Bank size (log)	44,722	7.636	1.970	2.014	8.675	9.134
CAR (%)	44,028	13.259	2.022	8.480	13.170	17.720
Z-score	42,671	5.413	0.909	3.338	5.390	7.471
NIM (%)	44,595	2.115	0.582	0.225	2.099	4.075
Loan-to-deposit (%)	44,594	71.796	15.761	33.163	73.046	111.223
Loan-to-asset (%)	44,459	49.486	8.955	25.112	51.812	64.152
Panel C: Loan-level analysis	N	Mean	SD	P1	Median	P99
Loan spread	7,474	26.625	106.298	-335	15	369
Loan amount (million RMB)	13,358	106.177	623.442	0.500	30	1,000
Firm size	13,358	8.653	1.15	6.347	8.542	11.716
Firm leverage (%)	13,358	51.253	18.436	10.709	51.239	93.989
Firm tangibility (%)	13,358	20.812	16.264	0.138	17.268	66.911
Firm cash holdings (%)	13,358	15.162	9.592	1.177	13.091	49.742
Firm ROA (%)	13,358	2.349	6.618	-38.937	2.807	14.848
Panel D: Prefecture- and province-level variables	N	Mean	SD	P1	Median	P99
Share of local banks	3,136	0.381	0.154	0.008	0.382	0.688
Credit-to-GDP (%)	3,136	99.935	57.851	30.199	82.423	335.777
GDP growth (%)	3,136	8.704	4.164	-4.8	8.4	18.2
Fiscal balance (%)	3,136	12.345	10.279	-0.083	9.787	56.609
SD of GDP growth	3,643	0.501	0.500	0	1	1
CV of GDP growth	3,643	0.501	0.500	0	1	1
Vacancy>6m	3,729	0.020	0.141	0	0	1
Turnover<3y	3,729	0.317	0.465	0	0	1
New hires (log)	387	3.169	0.824	1.099	3.401	4.419

Table 3: Penalties: Baseline results

The table presents difference-in-differences estimates of the effect of the 2015 decentralization reform on penalties based on the model in (2). Columns 1-4 present results using the likelihood of a penalty as dependent variable, while columns 5-8 present results using the number of penalties (log) as dependent variable. Observations are bank-prefecture-years from 2010 to 2020. All variables are defined in Appendix B. Robust standard errors are clustered by prefecture. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

	Penalty dummy				Number of penalties			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Local bank	-0.015*** (0.003)				-0.012*** (0.002)			
Post	0.105*** (0.007)				0.110*** (0.008)			
Local bank $\times$ Post	0.067*** (0.007)	0.075*** (0.008)	0.063*** (0.008)	0.088*** (0.008)	0.078*** (0.009)	0.088*** (0.010)	0.073*** (0.009)	0.104*** (0.010)
Year FE	No	Yes	No	Yes	No	Yes	No	Yes
Bank FE	No	Yes	Yes	No	No	Yes	Yes	No
Prefecture FE	No	Yes	No	No	No	Yes	No	No
Prefecture $\times$ Year FE	No	No	Yes	No	No	No	Yes	No
Bank $\times$ Prefecture FE	No	No	No	Yes	No	No	No	Yes
Observations	52,773	52,773	52,756	52,769	52,773	52,773	52,756	52,769
R ²	0.047	0.144	0.245	0.213	0.046	0.141	0.239	0.208

Table 4: Decentralization and penalties: Controls and differential trends

The table presents difference-in-differences estimates of the effect of the 2015 decentralization reform on penalties based on the model in (2) further controlling for bank and branch characteristics as well as differential trends. Odd-numbered (even-numbered) columns present results using the likelihood of a penalty (the number of penalties (log)) as dependent variable. Panel A (Panel B) includes controls for bank (branch) characteristics and differential trends based on these characteristics, with all characteristics measured at their average pre-2015 values. Observations are bank-prefecture-years from 2010 to 2020 for Panel A, while branch-years from 2012 to 2020 for Panel B. All variables are defined in Appendix B. Robust standard errors are clustered by prefecture. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Panel A: Bank characteristics	Penalty dummy	Number of penalties	Penalty dummy	Number of penalties
	(1)	(2)	(3)	(4)
Local bank $\times$ Post	0.023*** (0.009)	0.033*** (0.009)	0.037** (0.017)	0.065*** (0.019)
Bank size	0.012 (0.009)	0.004 (0.009)		
CAR	-0.004*** (0.002)	-0.006*** (0.002)		
Z-score	0.004 (0.002)	0.006** (0.003)		
NIM	0.009 (0.006)	0.003 (0.007)		
Loan-to-deposit	0.000 (0.000)	-0.000 (0.000)		
Loan-to-asset	-0.001 (0.001)	-0.000 (0.001)		
Bank size (pre-2015) $\times$ Post			0.005 (0.004)	0.009** (0.005)
CAR (pre-2015) $\times$ Post			0.001 (0.001)	0.002 (0.002)
Z-score (pre-2015) $\times$ Post			-0.032*** (0.009)	-0.033*** (0.010)
NIM (pre-2015) $\times$ Post			-0.002 (0.009)	0.003 (0.011)
Loan-to-deposit (pre-2015) $\times$ Post			-0.002* (0.001)	-0.002 (0.001)
Loan-to-asset (pre-2015) $\times$ Post			0.002* (0.001)	0.001 (0.001)
Year FE	Yes	Yes	Yes	Yes
Bank $\times$ Prefecture FE	Yes	Yes	Yes	Yes
Observations	42,556	42,556	42,098	42,098
R ²	0.219	0.214	0.215	0.211

Continued on next page...

Table 4: Penalties: Controls and differential trends (Continued)

Panel B: Branch characteristics	Penalty dummy (1)	Number of penalties (2)	Penalty dummy (3)	Number of penalties (4)
Local bank $\times$ Post	0.074*** (0.013)	0.106*** (0.014)	0.075*** (0.012)	0.107*** (0.014)
Branch size	-0.015*** (0.005)	-0.019*** (0.006)		
Branch ROA	-0.002 (0.002)	-0.003 (0.003)		
Branch size (pre-2015) $\times$ Post			0.043*** (0.006)	0.049*** (0.007)
Branch ROA (pre-2015) $\times$ Post			-0.000 (0.004)	0.004 (0.004)
Year FE	Yes	Yes	Yes	Yes
Bank $\times$ Prefecture FE	Yes	Yes	Yes	Yes
Observations	18,721	18,721	17,238	17,238
R ²	0.254	0.255	0.257	0.251

Table 5: Average fines and dispersion

The table presents difference-in-differences estimates of the effect of the 2015 decentralization reform on average fine amount and dispersion. Columns 1-2 present results using the average fine, in levels and logs, respectively, as the dependent variable. Columns 3-4 present results using the normalized absolute deviation of fines and the absolute deviation of log fines from the mean log fine, respectively, as the dependent variable. Observations are bank-prefecture-years with positive fines from 2010 to 2020. All variables are defined in Appendix B. Robust standard errors are clustered by prefecture. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

	Average fine (1)	Average fine (log) (2)	Fine dispersion (3)	Fine dispersion (log) (4)
Local bank $\times$ Post	-465.083* (276.252)	-0.272** (0.116)	0.107** (0.041)	0.100* (0.055)
Year FE	Yes	Yes	Yes	Yes
Bank $\times$ Prefecture FE	Yes	Yes	Yes	Yes
Observations	3,871	3,871	3,871	3,871
R ²	0.333	0.545	0.505	0.496

Table 6: Access to local information

The table presents difference-in-differences estimates of the effect of the 2015 decentralization reform on penalties based on the model in (2) further interacting with measures of geographic distance between the prefecture of the branch and Beijing. Odd-numbered (even-numbered) columns present results using the likelihood of a penalty (the number of penalties (log)) as dependent variable. Observations are bank-prefecture-years from 2010 to 2020. All variables are defined in Appendix B. Robust standard errors are clustered by prefecture. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

	Distance (log)				Distance dummies	
	Penalty dummy (1)	Number of penalties (2)	Penalty dummy (3)	Number of penalties (4)	Penalty dummy (5)	Number of penalties (6)
Local bank $\times$ Post	-0.118* (0.068)	-0.189** (0.074)				
Post $\times$ Distance	-0.019* (0.011)	-0.022 (0.014)				
Local bank $\times$ Post $\times$ Distance	0.030*** (0.010)	0.043*** (0.011)	0.022*** (0.005)	0.032*** (0.007)		
Local bank $\times$ Post $\times$ Long distance					0.094*** (0.014)	0.122*** (0.021)
Local bank $\times$ Post $\times$ Intermediate distance					0.094*** (0.010)	0.111*** (0.013)
Local bank $\times$ Post $\times$ Short distance					0.077*** (0.011)	0.079*** (0.014)
Bank FE	No	No	No	No	No	No
Year FE	Yes	Yes	No	No	Yes	Yes
Bank $\times$ Year FE	No	No	Yes	Yes	No	No
Prefecture $\times$ Year FE	No	No	Yes	Yes	No	No
Bank $\times$ Prefecture FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	52,769	52,769	43,064	43,064	52,769	52,769
R ²	0.213	0.209	0.379	0.370	0.213	0.209

Table 7: Local uncertainty

The table presents difference-in-differences estimates of the effect of the 2015 decentralization reform on penalties based on the model in (2) further interacting with measures of local uncertainty. Columns 1-2 present results using the likelihood of a penalty as dependent variable, while columns 3-4 present results using the number of penalties (log) as dependent variable. Panel A (Panel B) includes interacted controls for local economic (local political) uncertainty. Observations are bank-prefecture-years from 2010 to 2020. All variables are defined in Appendix B. Robust standard errors are clustered by prefecture. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Panel A: Economic uncertainty	Penalty dummy		Number of penalties	
	(1)	(2)	(3)	(4)
Local bank $\times$ Post	0.069*** (0.012)	0.067*** (0.012)	0.083*** (0.014)	0.080*** (0.014)
SD of GDP growth	0.025*** (0.009)		0.033*** (0.010)	
Local bank $\times$ SD of GDP growth	-0.028*** (0.010)		-0.039*** (0.013)	
Post $\times$ SD of GDP growth	-0.031** (0.014)		-0.034** (0.014)	
Local bank $\times$ Post $\times$ SD of GDP growth	0.039** (0.016)		0.041** (0.019)	
CV of GDP growth		0.019** (0.009)		0.031*** (0.010)
Local bank $\times$ CV of GDP growth		-0.021** (0.010)		-0.028** (0.013)
Post $\times$ CV of GDP growth		-0.021* (0.012)		-0.029** (0.013)
Local bank $\times$ Post $\times$ CV of GDP growth		0.042*** (0.014)		0.048*** (0.018)
Year FE	Yes	Yes	Yes	Yes
Bank $\times$ Prefecture FE	Yes	Yes	Yes	Yes
Observations	51,999	51,999	51,999	51,999
R ²	0.213	0.213	0.209	0.209
Panel B: Political uncertainty	Penalty dummy		Number of penalties	
	(1)	(2)	(3)	(4)
Local bank $\times$ Post	0.085*** (0.008)	0.077*** (0.010)	0.100*** (0.010)	0.091*** (0.012)
Vacancy>6m	0.029** (0.013)		0.025** (0.012)	
Local bank $\times$ Vacancy>6m	-0.026* (0.016)		-0.020 (0.019)	
Post $\times$ Vacancy>6m	-0.068*** (0.024)		-0.062*** (0.021)	
Local bank $\times$ Post $\times$ Vacancy>6m	0.101** (0.042)		0.133** (0.064)	
Turnover<3y		0.017** (0.007)		0.020*** (0.007)
Local bank $\times$ Turnover<3y		-0.017** (0.008)		-0.017** (0.008)
Post $\times$ Turnover<3y		-0.028** (0.011)		-0.028** (0.012)
Local bank $\times$ Post $\times$ Turnover<3y		0.033** (0.014)		0.035** (0.016)
Year FE	Yes	Yes	Yes	Yes
Bank $\times$ Prefecture FE	Yes	Yes	Yes	Yes
Observations	52,553	52,553	52,553	52,553
R ²	0.213	0.213	0.209	0.208

Table 8: Supervisory capacity

The table presents difference-in-differences estimates of the effect of the 2015 decentralization reform on penalties based on the model in (2) further controlling for local supervisory capacity, using either the full sample or a restricted sample ending in 2017 (i.e., prior to the 2018 CBRC–CBIRC merger). Odd-numbered (even-numbered) columns present results using the likelihood of a penalty (the number of penalties (log)) as dependent variable. Observations are bank-prefecture-years from 2010 to 2020 for the full sample and 2010-2017 in the restricted sample. All variables are defined in Appendix B. Robust standard errors are clustered by prefecture. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

	Full sample		Before 2018 merger	
	Penalty dummy (1)	Number of penalties (2)	Penalty dummy (3)	Number of penalties (4)
Local bank $\times$ Post	0.088*** (0.008)	0.104*** (0.010)	0.068*** (0.008)	0.075*** (0.009)
New hires	-0.004 (0.006)	-0.003 (0.006)	-0.002 (0.006)	0.001 (0.005)
Year FE	Yes	Yes	Yes	Yes
Bank $\times$ Prefecture FE	Yes	Yes	Yes	Yes
Observations	52,769	52,769	41,856	41,856
R ²	0.213	0.208	0.240	0.233

Table 9: Lending: Loan-level analysis

The table presents difference-in-differences estimates of the effect of the 2015 decentralization reform on loan conditions based on the model similar to (2). Columns 1-2 present results using loan spreads as dependent variable, while columns 3-4 present results using loan amounts (log) as dependent variable. Observations are loan-branch-years from 2010 to 2020. All variables are defined in Appendix B. Robust standard errors are clustered by firm. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

	Loan spread		Loan amount	
	(1)	(2)	(3)	(4)
Local bank $\times$ Post	31.477** (13.315)	31.257** (12.448)	-0.385*** (0.139)	-0.329** (0.139)
Firm size	4.595 (6.432)	6.563 (5.934)	-0.022 (0.074)	-0.051 (0.059)
Firm leverage	0.278 (0.304)	0.102 (0.366)	0.002 (0.002)	0.002 (0.002)
Firm tangibility	-0.388** (0.193)	-0.523*** (0.183)	0.001 (0.003)	0.000 (0.002)
Firm cash holdings	0.042 (0.344)	0.095 (0.406)	0.002 (0.003)	-0.001 (0.003)
Firm ROA	0.040 (0.356)	-0.373 (0.371)	0.001 (0.004)	0.001 (0.003)
Year FE	Yes	Yes	Yes	Yes
Bank FE	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes
Prefecture FE	No	Yes	No	Yes
Observations	7,229	7,203	13,012	12,986
R ²	0.602	0.657	0.396	0.458

Internet Appendix to

“Subordinates in Charge: Does Delegation Improve Bank Supervision?”

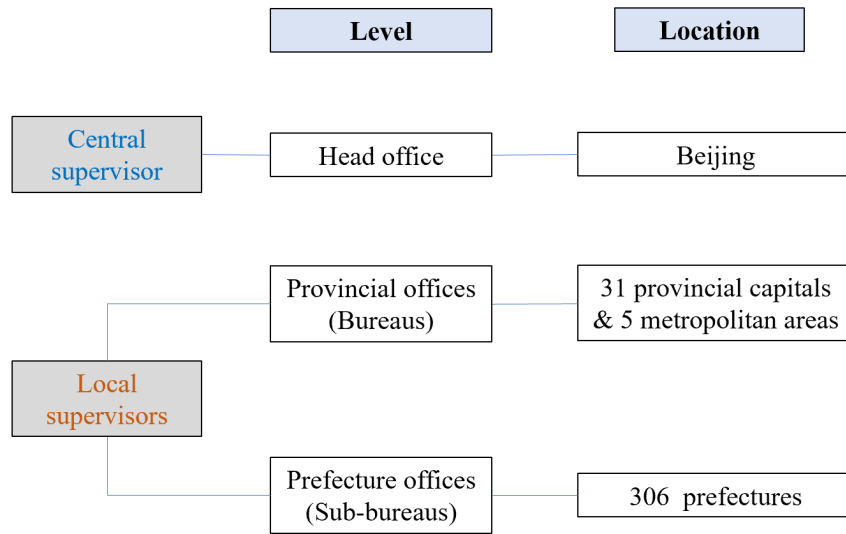


Figure IA1: CBRC Structure

Information disclosure form for supervisory penalty, Yichang sub-bureau of the CBRC

宜昌银监分局行政处罚信息公开表
(湖北银行股份有限公司宜昌分行)

Penalty ID	行政处罚决定书文号		宜银监罚决字[2015]3号	No. 3, Yichang authority 2015
Punished entity	被处罚当事人姓名或名称	个人姓名 Individual	---	Yichang branch, Bank of Hubei
	单位名称	名称 Bank name	湖北银行股份有限公司宜昌分行 法定代表人（主要负责人）姓名 Bank CEO 何青平	
Facts of misconduct	主要违法违规事实（案由）		存在贷款风险分类不准确、以贷转存吸收存款的违规行为。	Misconducts of inaccurately classifying loans based on their risk level and withholding a certain percentage of the loan as a deposit when the loan is granted.
Penalty basis	行政处罚依据		《中华人民共和国银行业监督管理法》第四十六条第（五）项	Banking Supervision and Administration Law of the PRC, Article 46(5).
Penalty decision	行政处罚决定		罚款人民币40万元	A fine of 400,000 RMB
Supervisory office	作出处罚决定的机关名称		中国银行业监督管理委员会宜昌监管分局	Yichang sub-bureau of the CBRC
Decision date	作出处罚决定的日期		2015年11月5日	November 5, 2015

Figure IA2: Example of a penalty

The figure presents a snapshot of a randomly chosen penalty notice published on the CBIRC website (source: www.cbirc.gov.cn; last accessed: August 2023).

Table IA1: Branches at different levels of aggregation

The table presents the composition of national and local banks in our sample and provides summary statistics on key characteristics at the branch, bank, and aggregate levels. All variables are defined in Appendix B.

Aggregation level	Local banks	National banks
Branch		
Total assets (log million RMB)	11.76	11.79
ROA	0.97%	0.98%
Market share (offices)	5.03%	6.56%
Distance to Beijing (kilometers)	1101.98	1183.73
Credit to GDP	108.04%	112.79%
HHI of branches' prefecture (offices)	0.09	0.10
Number of penalties	2.80	1.86
Bank		
Number of branches	2.03	182.56
Number of penalties	5.68	337.78
All		
Number of banks	1056	18
Number of penalties	5,998	6,080
Total fine amount (million)	1761.7	3690.69

Table IA2: Penalties: Subsample analysis

The table presents difference-in-differences estimates of the effect of the 2015 decentralization reform on penalties based on the model in (2) for different subsample of banks. Columns 1-2 present results using either the likelihood of a penalty or the number of penalties (log) as dependent variable and excluding from the sample state-owned banks. Columns 3-4 present results using either the likelihood of a penalty or the number of penalties (log) as dependent variable and excluding from the sample penalties issued by the local offices of the CBRC in Beijing. Observations are bank-prefecture-years from 2010 to 2020. All variables are defined in Appendix B. Robust standard errors are clustered by prefecture. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

	Excluding state-owned banks		Excluding penalties in Beijing	
	Penalty dummy (1)	Number of penalties (2)	Penalty dummy (3)	Number of penalties (4)
Local bank $\times$ Post	0.063*** (0.012)	0.086*** (0.014)	0.090*** (0.008)	0.106*** (0.010)
Year FE	Yes	Yes	Yes	Yes
Bank $\times$ Prefecture FE	Yes	Yes	Yes	Yes
Observations	31,914	31,914	52,439	52,439
R ²	0.223	0.217	0.211	0.206

Table IA3: Penalties (type and recipient): Poisson regressions

The table presents difference-in-differences estimates of the effect of the 2015 decentralization reform on penalties (type and recipient) based on the model in (2) using Poisson regressions. Columns 1-7 present results using the number of each type or recipient of penalties as dependent variable and implementing Poisson pseudo maximum likelihood regressions with (multiple levels of) fixed effects as described by [Correia et al. \(2020\)](#). The number of license revocations is not reported because there are too few observations of this penalty type. Observations are bank-prefecture-years from 2010 to 2020. All variables are defined in Appendix B. Robust standard errors are clustered by prefecture. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

	Number of penalties	Number of fines	Number of warnings	Number of disqualifications	Number of prohibitions	Number of individual penalties	Number of bank penalties
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Local bank $\times$ Post	1.173*** (0.141)	0.990*** (0.112)	1.537*** (0.279)	1.268*** (0.406)	1.518 (1.186)	1.162*** (0.381)	1.032*** (0.130)
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Bank $\times$ Prefecture FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	33,906	33,021	15,573	2,515	2,218	14,790	32,720
Pseudo R ²	0.321	0.265	0.350	0.136	0.299	0.392	0.242

Table IA4: Penalties (type and recipient)

The table presents difference-in-differences estimates of the effect of the 2015 decentralization reform on penalties (type and recipient) based on the model in (2). Panel A presents results on each type of penalties, while Panel B presents results on each recipient of penalties. Odd-numbered (even-numbered) columns present results using the likelihood of a penalty (the number of penalties (log)) as dependent variable. License revocations are not reported because there are too few observations of this penalty type. Observations are bank-prefecture-years from 2010 to 2020. All variables are defined in Appendix B. Robust standard errors are clustered by prefecture. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Panel A: Type of penalties	Fine dummy (1)	Number of fines (2)	Warning dummy (3)	Number of warnings (4)	Disqualification dummy (5)	Number of disqualifications (6)	Prohibition dummy (7)	Number of prohibitions (8)
Local bank $\times$ Post	0.084*** (0.008)	0.079*** (0.008)	0.049*** (0.005)	0.057*** (0.007)	0.007*** (0.002)	0.006*** (0.001)	0.004*** (0.002)	0.004*** (0.001)
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Bank $\times$ Prefecture FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	52,769	52,769	52,769	52,769	52,769	52,769	52,769	52,769
R ²	0.212	0.212	0.148	0.149	0.102	0.104	0.110	0.110
Panel B: Recipient of penalties	Individual penalty dummy (1)	Number of individual penalties (2)	Bank penalty dummy (3)	Number of bank penalties (4)				
Local bank $\times$ Post	0.049*** (0.005)	0.060*** (0.007)	0.081*** (0.008)	0.069*** (0.007)				
Year FE	Yes	Yes	Yes	Yes				
Bank $\times$ Prefecture FE	Yes	Yes	Yes	Yes				
Observations	52,769	52,769	52,769	52,769				
R ²	0.155	0.153	0.210	0.212				

Table IA5: Penalties: Dynamic effects

The table presents difference-in-differences estimates of the effect of the 2015 decentralization reform on penalties based on the model in (2) with $Local\ bank_i$ interacted with $Year_{(2015-[+])1}$, that is, each year before and after 2015 (excluding 2015). Column 1 presents results using the likelihood of a penalty as dependent variable, while column 2 presents results using the number of penalties (log) as dependent variable. Observations are bank-prefecture-years from 2010 to 2020. All variables are defined in Appendix B. Robust standard errors are clustered by prefecture. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

	Penalty dummy (1)	Number of penalties (2)
Local bank $\times$ Year 2010	-0.000 (0.015)	-0.005 (0.013)
Local bank $\times$ Year 2011	-0.000 (0.014)	-0.005 (0.013)
Local bank $\times$ Year 2012	0.019 (0.014)	0.011 (0.013)
Local bank $\times$ Year 2013	0.019 (0.014)	0.015 (0.013)
Local bank $\times$ Year 2014	0.012 (0.015)	0.011 (0.013)
Local bank $\times$ Year 2016	0.053*** (0.014)	0.050*** (0.013)
Local bank $\times$ Year 2017	0.120*** (0.016)	0.129*** (0.018)
Local bank $\times$ Year 2018	0.158*** (0.018)	0.167*** (0.018)
Local bank $\times$ Year 2019	0.114*** (0.015)	0.132*** (0.017)
Local bank $\times$ Year 2020	0.149*** (0.019)	0.180*** (0.023)
Year FE	Yes	Yes
Bank $\times$ Prefecture FE	Yes	Yes
Observations	52,769	52,769
R ²	0.217	0.213

Table IA6: Lending: Prefecture-level analysis

The table presents difference-in-differences estimates of the effect of the 2015 decentralization reform on loan supply at the prefecture level based on model similar to (2) using either OLS or 2SLS regressions. Columns 1-2 present OLS results using credit to GDP at the prefecture level as dependent variable, while columns 3-4 present 2SLS results using the predetermined share of local banks in 2010 as an instrument, following [Gilje et al. \(2016\)](#). Observations are prefecture-years from 2010 to 2020. All variables are defined in Appendix B. Robust standard errors are clustered by prefecture. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

	Credit to GDP			
	OLS		2SLS	
	(1)	(2)	(3)	(4)
Local bank share	-70.729*** (17.093)	-41.114** (18.554)	-73.159*** (17.874)	-35.427* (20.028)
Local bank share $\times$ Post	-31.245** (15.558)	-35.624*** (12.240)	-43.634*** (16.437)	-32.137** (14.063)
GDP growth	-0.427 (0.706)	-0.561 (0.349)	-0.387 (0.706)	-0.572 (0.350)
Fiscal balance	-0.122 (0.303)	-0.792** (0.381)	-0.100 (0.302)	-0.824** (0.397)
Year FE	Yes	Yes	Yes	Yes
Province FE	No	Yes	No	Yes
Observations	3,136	3,136	3,136	3,136
R ²	0.153	0.440	0.061	0.060
First-stage F-statistic			201.85	195.47